MOKO

MOKO Bluetooth Gateway

Remote Management Commands

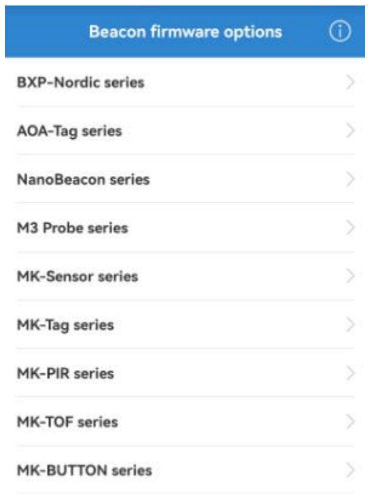
Version V2.3

Contents

1. About This Manual

This document describes the remote management commands which are sent from cloud to gateway to connect and manage MOKO Beacon devices via the gateway.

MOKO offers a wide range of Bluetooth Beacon products, and multiple models can run the same firmware. Therefore, the management commands for MOKO beacons on the gateway are defined based on Beacon firmware type. Below are the connection types supported in the gateway firmware and the defined Beacon firmware types:

Connection type in gateway firmware	Beacon firmware type	BeaconX Pro APP user interface
BXP-B-CR	MK-Button	
BXP-B-D		
BXP-C	BXP-Nordic series	
BXP-D		
BXP-T	MK-Tag	
BXP-S	MK-Sensor	
MK PIR	MK-PIR	
MK TOF	MK-TOF	

If the firmware type of your Beacon device is not clear, especially for the MK-Button and BXP-Nordic devices, the MOKO BeaconX Pro mobile APP could help to check this information. Please see the Appendix to get details.

This document applies to MOKO MKGW3, MKGW-mini 02, MK107Pro **V2.X.X** firmware and MKGW8 gateways.

2. Common Commands

Below commands are suitable for all MOKO beacon types.

2.1 Disconnect from BLE device (200)

Send command from cloud to the gateway, to make the gateway disconnect from the BLE device.

Flag	Msg id offset	Parameter name	Parameter value
Write	200	mac	string. 12 characters (6 bytes of BLE device mac address)

Setup:


```
{
	"msg_id": 1200,
	"device_info": {
		"mac": "4c11ae8be624"
	},
	"data": {
		"mac": "f505ecb011dd"
	}
}
```


Reply:


```
{
	"msg_id": 1200,
	"device_info": {
		"mac": "4c11ae8be624"
	},
	"result_code": 0,
	"result_msg": "success"
}
```


2.2 Get gateway connection status (201)

To get the gateway and BLE device connection status.

Flag	Msg id offset	Parameter name	Parameter value
Read	201	ble_conn_list	Array. The connection list
		mac	String. BLE device mac address connected with gateway
		type	number

			0: common type 1: BXP-B-D 2: BXP-CR 3: BXP-C 4: BXP-D 5: BXP-T 6: BXP-S 7: PIR 8: TOF

Setup:


```
{
	"msg_id": 2201,
	"device_info": {
		"mac": "4c11ae8be624"
	}
}
```


Reply:


```
{
  "msg_id" : 2201,
  "device_info" : {
    "mac" : "4c11ae8be624"
  },
  "data" : {
    "ble_conn_list" : [ {
      "mac" : "f505ecb011dd",
      "type" : 4
    } ]
  }
}
```


2.3 Upgrade Beacon firmware (205)

To remotely upgrade BLE device firmware via gateway. Max 20 BLE devices could be included in one command.

Flag	Msg id offset	Parameter name	Parameter value
write	205	beacon_type	number 1: BXP-B-D 2: BXP-CR 3: BXP-C 4: BXP-D

			5: BXP-T 6: BXP-S 7: PIR 8: TOF
		firmware_url	String. 1-256 characters
		init_data_url	String. 1-256 characters
		ble_dev	Array. At most 20 BLE devices could be included.
		mac	String. 12 characters, mac address of BLE device
		passwd	String. 0-16 charaxters, password of BLE device

Setup:


```
{
   "msg_id" : 1205,
   "device_info" : {
      "mac" : "4c11ae8be624"
   },
   "data" : {
      "beacon_type" : 4,
      "firmware_url" : "http://47.104.172.169:8080/updata_fold/bxp_d.bin",
      "init_data_url" : "http://47.104.172.169:8080/updata_fold/bxp_d_data.bin",
      "ble_dev" : [ {
         "mac" : "f505ecb011dd",
         "passwd" : ""
      } ]
   }
}
```


Reply:


```
{
      "msg_id": 1205,
      "device_info": {
            "mac": "4c11ae8be624"
      },
      "result_code": 0,
      "result_msg": "success"
}
```

2.4 Report Beacon firmware update process (206)

After the gateway receives the upgrade Beacon firmware command, it will connect the devices and update their firmware one bye one. The gateway will report the upgrade percent for each BLE device during upgrading.

Flag	Msg id offset	Parameter name	Parameter value
Notify	206	mac	string. 12 characters (6 bytes of BLE device mac address)
		percent	number. 0-100 Firmware update process in percent.


```
{
  "msg_id" : 3206,
  "device_info" : {
    "mac" : "142b2fe9b19c"
  },
  "data" : {
    "mac" : "f505ecb011dd",
    "percent" : 3
  }
}
```


2.5 Report Beacon firmware update result (207)

The gateway will report the result for each device after firmware update is complete.

Flag	Msg id offset	Parameter name	Parameter value
Notify	207	mac	string. 12 characters (6 bytes of BLE device mac address)
		status	number. 0: upgrade start 1: upgrade success 2: upgrade fail


```
{
    "msg_id": 3207,
    "device_info": {
        "mac": "4c11ae8be624"
    },
    "data": {
```

```
        "mac": "f505ecb011dd",
        "status": 1
    }
}
```

2.6 Report batch firmware update result(208)

The gateway will report the result after all BLE devices firmware upgrade is complete.

Flag	Msg id offset	Parameter name	Parameter value
Notify	208	multi_dfu_reasult_code	0: could not find the firmware files 1: batch upgrade is complete
		fail_dev	array. Failure device list
		mac	string. Failure device mac address
		reason	number. 0: BLE device is not in range 1: connect to ble device failed 2: password error 3: upgrade is failed 4: beacon type error

If the gateway could not find the firmware URLs:

```
{
  "msg_id" : 3208,
  "device_info" : {
    "mac" : "142b2fe9b19c"
  },
  "data" : {
    "multi_dfu_result_code" : 0
  }
}
```

After the batch upgrade is complete:

```
{
  "msg_id" : 3208,
  "device_info" : {
    "mac" : "142b2fe9b19c"
  },
  "data" : {
    "multi_dfu_result_code" : 1,
    "fail_dev" : [ {
```

```
        "mac" : "f505ecb011dd",
        "reason" : 2
      } ]
    }
}
```

2.7 Communication timeout (209)

If the gateway and BLE device have no any communication for a while, gateway will automatically disconnect from the BLE device.
This command is to set or get the communication timeout.

Flag	Msg id offset	Parameter name	Parameter value
Read/write	209	timeout	Number 0-60, unit: minute Set to 0: automatic disconnection will not perform

Inquire:

```
{
    "msg_id": 2209,
    "device_info": {
        "mac": "4c11ae8be624"
    }
}
```

Reply:

```
{
    "msg_id": 2209,
    "device_info": {
        "mac": "4c11ae8be624"
    },
    "data": {
        "timeout": 10
    }
}
```

Setup:

```
{
    "msg_id": 1209,
    "device_info": {
        "mac": "4c11ae8be624"
    },
    "data": {
```

```
        "timeout": 10
    }
}
```

Reply:

```
{
    "msg_id": 1209,
    "device_info": {
        "mac": "4c11ae8be624"
    },
    "result_code": 0,
    "result_msg": "success"
}
```

2.8 Report disconnection type (210)

When the gateway disconnects from the BLE device due to not a non-manual operation, it will report a message to cloud.

Flag	Msg id offset	Parameter name	Parameter value
Notify	210	mac	String. BLE device mac address
		reason	number 0: abnormal disconnection 1: communication timeout disconnection


```
{
  "msg_id" : 3210,
  "device_info" : {
    "mac" : "142b2fe9b19c"
  },
  "data" : {
    "mac" : "f505ecb011dd",
    "reason" : 0
  }
}
```

3. Connect and Manage BXP-B-D device

Send command from cloud to the gateway, to make the gateway connect and communicate with the MK Button device.
The MK button devices has two firmware branches: BXP-B-D and BXP-B-CR. This chapter describes the remote commands for BXP-B-D devices.

3.1 Connect MOKO Button device (100)

Send command from cloud to the gateway, to make the gateway connect with the MK Button device.

Flag	Msg id offset	Parameter name	Parameter value
Write	100	mac	string. 12 characters (6 bytes of BLE device mac address)
		passwd	string. 0-16 characters (0-8 bytes of password).

Setup:


```
{
    "msg_id": 1100,
    "device_info": {
        "mac": "4c11ae8be624"
    },
    "data": {
        "mac": "D9A5FA6F3882",
        "passwd": "Moko4321"
    }
}
```


Reply:


```
{
    "msg_id": 1100,
    "device_info": {
        "mac": "4c11ae8be624"
    },
    "result_code": 0,
    "result_msg": "success"
}
```


3.2 Report connection result (101)

After the gateway finish the connection, it will actively report the connection result to cloud. If connecting with MK button device successfully, the gateway will also inquire the device information from the button device and transfer to cloud.

Flag	Msg id offset	Parameter name	Parameter value
Notify	101	mac	string. 12 characters (6 bytes of BLE device mac address)
		result_code	number. 0-8 0: Connect succeed 1: Gateway BLE scan disabled 2: Exceed the connectable number (the gateway is connected with an another BLE device) 3: BLE device connected (the BLE device has been connected with gateway) 4: BLE device out of range 5: BLE device un-connectable 6: Connect fail 7: BLE device type error(connection type error) 8: Password error
		result_msg	string. The description for the result code
		product_model	string. Valid when result code is 0
		company_name	string. Valid when result code is 0
		hardware_version	string. Valid when result code is 0
		software_version	string. Valid when result code is 0
		firmware_version	string. Valid when result code is 0
		battery_v	number. Battery voltage, valid when result code is 0
		battery_level	**number.** **Battery percentage, valid when result code is 0** **Added in gateway V2.X.X firmware.**
		single_alarm_num	number. Single press alarm number, valid

			when result code is 0
		double_alarm_num	number. Double press alarm number, valid when result code is 0
		long_alarm_num	number. Long press alarm number, valid when result code is 0
		sensor_status	number. Valid when result code is 0. ➢ Bit0: 3-axis sensor equipped status. 0- not equipped, 1- equipped. ➢ Bit1: temperature&humidity sensor equipped status. 0- not equipped, 1- equipped. ➢ Bit2: light sensor equipped status. 0- not equipped, 1- equipped.
		alarm_status	number. Valid when result code is 0. ➢ Bit0: single press alarm triggered status. 0- not trigger, 1- triggered. ➢ Bit1: double press alarm triggered status. 0- not trigger, 1- triggered. ➢ Bit2: long press alarm triggered status. 0- not trigger, 1- triggered. ➢ Bit3: Abnormal inactivity triggered status. 0- not trigger, 1- triggered.


```
{
    "msg_id": 3101,
    "device_info": {
        "mac": "4c11ae8be624"
    },
    "data": {
     "mac" : "fdcbe50c4987",
    "result_code" : 0,
    "result_msg" : "Connect succeed",
    "product_model" : "BUTTON",
    "company_name" : "MOKO TECHNOLOGY LTD.",
```

```
    "hardware_version" : "MKBN Series",
    "software_version" : "BXP-B-D",
    "firmware_version" : "V2.0.3",
    "sensor_status" : 1,
    "battery_v" : 3003,
    "battery_level" : 80,
    "single_alarm_num" : 0,
    "double_alarm_num" : 0,
    "long_alarm_num" : 0,
    "alarm_status" : 0
  }
}
```

3.3 Get BLE device information (102)

Send command from cloud to the gateway, to make the gateway inquire the device information of the MK Button device.

Flag	Msg id offset	Parameter name	Parameter value
Write	102	mac	string. 12 characters (6 bytes of BLE device mac address)

Setup:

```
{
    "msg_id": 1102,
    "device_info": {
        "mac": "4c11ae8be624"
    },
    "data": {
        "mac": "D9A5FA6F3882"
    }
}
```

Reply:

```
{
    "msg_id": 1102,
    "device_info": {
        "mac": "4c11ae8be624"
    },
    "result_code": 0,
    "result_msg": "success"
}
```

3.4 Report BLE device information (103)

After the gateway gets device information from the MK Button device, it will transfer to cloud.

Flag	Msg id offset	Parameter name	Parameter value
Notify	103	mac	string. 12 characters (6 bytes of BLE device mac address)
		result_code	number. 0-2 0: command success 1: the Button is not connected by gateway 2: command is not supported
		product_model	string. Valid when the result code is 0
		company_name	string. Valid when the result code is 0
		hardware_version	string. Valid when the result code is 0
		software_version	string. Valid when the result code is 0
		firmware_version	string. Valid when the result code is 0
		sensor_status	number. Valid when result code is 0. ➢ Bit0: 3-axis sensor equipped status. 0- not equipped, 1- equipped. ➢ Bit1: temperature&humidity sensor equipped status. 0- not equipped, 1- equipped. ➢ Bit2: light sensor equipped status. 0- not equipped, 1- equipped.


```
{
    "msg_id": 3103,
    "device_info": {
        "mac": "4c11ae8be624"
    },
    "data": {
     "mac" : "fdcbe50c4987",
    "result_code" : 0,
    "result_msg" : "Connect succeed",
```

```
        "product_model"  :  "BUTTON",
        "company_name"  :  "MOKO  TECHNOLOGY  LTD.",
        "hardware_version"  :  "MKBN  Series",
        "software_version"  :  "BXP-B-D",
        "firmware_version"  :  "V2.0.3",
        "sensor_status"  :  1,
        }
}
```


3.5 Get BLE device status (104)

To inquire the alarm status and battery status of the MK Button device.

Flag	Msg id offset	Parameter name	Parameter value
Write	104	mac	string. 12 characters (6 bytes of BLE device mac address)

Setup:


```
{
        "msg_id":  1104,
        "device_info":  {
                "mac":  "4c11ae8be624"
        },
        "data":  {
                "mac":  "D9A5FA6F3882"
        }
}
```


Reply:


```
{
        "msg_id":  1104,
        "device_info":  {
                "mac":  "4c11ae8be624"
        },
        "result_code":  0,
        "result_msg":  "success"
}
```


3.6 Report BLE device status (105)

After the gateway gets device status from the MK-Button device, it will transfer the reply to cloud.

Flag	Msg id offset	Parameter name	Parameter value

Notify	105	mac	string. 12 characters (6 bytes of BLE device mac address)
		result_code	number. 0-3 0: command success 1: the BLE device is not connected by gateway 2: command is not supported 3: command timeout
		battery_v	number. Battery voltage, valid when the result code is 0
		battery_level	**number.** **Battery percentage, valid when result code is 0** **Added in gateway V2.X.X firmware.**
		single_alarm_num	number. single press alarm number. Valid when the result code is 0
		double_alarm_num	number. double press alarm number. Valid when the result code is 0
		long_alarm_num	number. long press alarm number. Valid when the result code is 0
		alarm_status	number. Valid when result code is 0. ➢ Bit0: single press alarm triggered status. 0- not trigger, 1- triggered. ➢ Bit1: double press alarm triggered status. 0- not trigger, 1- triggered. ➢ Bit2: long press alarm triggered status. 0- not trigger, 1- triggered. Bit3: Abnormal inactivity triggered status. 0- not trigger, 1- triggered.


```
{
  "msg_id" : 3105,
  "device_info" : {
    "mac" : "142b2fe9b19c"
  },
```

"data" : {
"mac" : "fdcbe50c4987",
"result_code" : 0,
"battery_v" : 3003,
"battery_level" : 80,
"single_alarm_num" : 0,
"double_alarm_num" : 0,
"long_alarm_num" : 0,
"alarm_status" : 0
}
}

3.7 Dismiss alarm status (106)

Send this command to the gateway to dismiss the alarm status of the MK Button device.

Flag	Msg id offset	Parameter name	Parameter value
Write	106	mac	string. 12 characters (6 bytes of BLE device mac address)

Setup:

{
"msg_id": 1106,
"device_info": {
"mac": "4c11ae8be624"
},
"data": {
"mac": "D9A5FA6F3882"
}
}

Reply:

{
"msg_id": 1106,
"device_info": {
"mac": "4c11ae8be624"
},
"result_code": 0,
"result_msg": "success"
}

3.8 Report alarm dismiss result (107)

After sending the dismiss alarm command, if the gateway gets reply from the MK Button device, it will transfer the reply to cloud.

Flag	Msg id offset	Parameter name	Parameter value
Notify	107	mac	string. 12 characters (6 bytes of BLE device mac address)
		result_code	number. 0-3 0: command success 1: the BLE device is not connected by gateway 2: command is not supported 3: command timeout


```
{
	"msg_id": 3107,
	"device_info": {
		"mac": "4c11ae8be624"
	},
	"data": {
		"mac": "d9a5fa6f3882",
		"result_code": 0
	}
}
```


3.9 Setup LED (109)

To setup the LED of the MK Button device.

Flag	Msg id offset	Parameter name	Parameter value
Write	109	mac	string. 12 characters (6 bytes of BLE device mac address)
		flash_time	LED flash duration. 1-6000, unit: 100 ms
		flash_interval	LED flash interval. 0-100, unit: 100 ms

Setup:


```
{
	"msg_id": 1109,
```

```
    "device_info": {
        "mac": "4c11ae8be624"
    },
    "data": {
        "mac": "D9A5FA6F3882",
        "flash_time":100,
        "flash_interval":10
    }
}
```

Reply:

```
{
    "msg_id": 1109,
    "device_info": {
        "mac": "4c11ae8be624"
    },
    "result_code": 0,
    "result_msg": "success"
}
```

3.10 Report LED setup result (110)

When the gateway gets reply from the MK Button Beacon for the setup LED command, it will report the result to cloud.

Flag	Msg id offset	Parameter name	Parameter value
Notify	110	mac	string. 12 characters (6 bytes of BLE device mac address)
		result_code	number. 0-3 0: command success 1: the BLE device is not connected by gateway 2: command is not supported 3: command timeout

```
{
    "msg_id": 3110,
    "device_info": {
        "mac": "4c11ae8be624"
    },
    "data": {
        "mac": "d9a5fa6f3882",
        "result_code": 0
```

```
    }
}
```

3.11 Setup Buzzer (111)

To setup the Buzzer of the MK Button device.

Flag	Msg id offset	Parameter name	Parameter value
Write	111	mac	string. 12 characters (6 bytes of BBLE device mac address)
		ring_time	Buzzer ring duration. 1-6000, unit: 100ms
		ring_interval	Buzzer ring interval. 0-100, unit: 100 ms

Setup:

```
{
    "msg_id": 1111,
    "device_info": {
        "mac": "4c11ae8be624"
    },
    "data": {
        "mac": "D9A5FA6F3882",
        "ring_time":100,
        "ring_interval":10
    }
}
```

Reply:

```
{
    "msg_id": 1111,
    "device_info": {
        "mac": "4c11ae8be624"
    },
    "result_code": 0,
    "result_msg": "success"
}
```

3.12 Report Buzzer setup result (112)

When the gateway gets reply from the MK Button device for the setup buzzer command, it will report the result to cloud.

Flag	Msg id offset	Parameter name	Parameter value
Notify	112	mac	string. 12 characters (6 bytes of BLE device mac address)
		result_code	number. 0-3 0: command success 1: the BLE device is not connected by gateway 2: command is not supported 3: command timeout


```
{
	"msg_id": 3112,
	"device_info": {
		"mac": "4c11ae8be624"
	},
	"data": {
		"mac": "d9a5fa6f3882",
		"result_code": 0
	}
}
```


3.13 Delete alarm trigger records(113)

Send command from cloud to gateway, to make the gateway delete the alarm trigger records from MK button device.

Flag	Msg id offset	Parameter name	Parameter value
Write	113	mac	string. 12 characters (6 bytes of BLE device mac address)
		type	number. 0-2 0: delete single press records 1: delete double click records 2: delete long press records

Setup:


```
{
	"msg_id": 1113,
	"device_info": {
		"mac": "4c11ae8be624"
	},
	"data": {
		"mac": "d9a5fa6f3882",
```

```
        "type": 0
    }
}
```

Reply:

```
{
    "msg_id": 1113,
    "device_info": {
        "mac": "4c11ae8be624"
    },
    "result_code": 0,
    "result_msg": "success"
}
```

3.14 Report alarm trigger records delete result (114)

When the gateway gets a reply from the MK Button device for the deleting trigger records command, it will report the result to cloud.

Flag	Msg id offset	Parameter name	Parameter value
Notify	114	mac	string. 12 characters (6 bytes of BLE device mac address)
		type	number. 0~2 0: delete single press record 1: delete double click record 2: delete long press record
		result_code	number. 0-3 0: command success 1: the BLE device is not connected by gateway 2: command is not supported 3: command timeout

```
{
    "msg_id": 3114,
    "device_info": {
        "mac": "4c11ae8be624"
    },
    "data": {
        "mac": "d9a5fa6f3882",
        "type": 0
        "result_code": 0
    }
}
```

3.15 Set 3-axis data notify switch (115)

To setup the MK-button to enable/ disable 3-axis data notify switch.

Flag	Msg id offset	Parameter name	Parameter value
Write	115	switch_value	number. 0-1 0: disable 1: enable
		mac	The mac of BLE device connected by gateway

Setup:


```
{
    "msg_id": 1115,
    "device_info": {
        "mac": "4c11ae8be624"
    }
   "data": {
        "mac": "d9a5fa6f3882",
        "switch_value": 0
    }

}
```


Reply:


```
{
    "msg_id": 1115,
    "device_info": {
        "mac": "4c11ae8be624"
    },
    "result_code": 0,
    "result_msg": "success"
}
```


3.16 Report 3-axis data notify switch set result (116)

When the gateway gets a reply from the MK Button device for the 3-axis notify switch setup command, it will report the result to cloud.

Flag	Msg id offset	Parameter name	Parameter value
Notify	116	mac	string. 12 characters BLE device MAC address
		result_code	number. 0-3

			0: command success 1: the BLE device is not connected with gateway 2: command is not supported 3: command timeout


```
{
    "msg_id": 3116,
    "device_info": {
        "mac": "4c11ae8be624"
    },
    "data": {
        "mac": "D9A5FA6F3882",
        "result_code": 0
    }
}
```


3.17 Report 3-axis data (117)

When the gateway enables the 3-axis data notify switch, it will receive real time 3-axis data from the MK Button device, then the gateway will transfer 3-axis data to cloud.

Flag	Msg id offset	Parameter name	Parameter value
Notify	117	mac	string. 12 characters BLE device MAC address
		x_axis_data	number, unit: mg
		y_axis_data	number, unit: mg
		z_axis_data	number, unit: mg


```
{
    "msg_id": 3117,
    "device_info": {
        "mac": "4c11ae8be624"
    },
    "data": {
        "mac": "d9a5fa6f3882",
        "x_axis_data": 20,
        "y_axis_data": 30,
        "z_axis_data": 20
    }
}
```

3.18 Set device power off (118)

To power off the beacon via gateway.

Flag	Msg id offset	Parameter name	Parameter value
Write	118	mac	string. 12 characters (6 bytes of BLE device mac address)

Setup:


```
{
      "msg_id": 1118,
      "device_info": {
            "mac": "4c11ae8be624"
      },
      "data": {
            "mac": "d9a5fa6f3882"
               }
}
```


3.19 Report set device power off result (119)

When the gateway gets a reply from the MK Button device for the power off setup command, it will transfer the result to cloud.

Flag	Msg id offset	Parameter name	Parameter value
Notify	119	result_code	number. 0-3 0: command success 1: the Button is not connected by gateway 2: command is not supported 3: command timeout
		mac	string. 12 characters (6 bytes of BLE device mac address)

Setup:


```
{
      "msg_id": 3119,
      "device_info": {
            "mac": "4c11ae8be624"
      },
      "data": {
            "mac": "d9a5fa6f3882",
            "result_code": 0
```

```
        }
}
```

3.20 Get broadcast parameters (120)

To inquire the Bluetooth broadcast parameters of MK Button device.

Flag	Msg id offset	Parameter name	Parameter value
Write	120	mac	string. 12 characters (6 bytes of BLE device mac address)

Setup:


```
{
    "msg_id": 1120,
    "device_info": {
        "mac": "4c11ae8be624"
    },
    "data": {
        "mac": "d9a5fa6f3882"
        }
}
```


3.21 Report broadcast parameters (121)

When the gateway gets reply from the MK Button device for the getting Bluetooth broadcast parameters command, it will transfer the result to cloud.

Flag	Msg id offset	Parameter name	Parameter value
Notify	121	mac	string. 12 characters (6 bytes of BLE device mac address)
		result_code	number. 0-3 0: command success 1: the BLE device is not connected by gateway 2: command is not supported 3: command timeout
		adv_param	Array. valid when result code is 0. The BLE device has four adv channels, therefore it has at most 4 groups of adv parameters.
		channel	number. 0~3

		enable	number. 0: disable 1: enable
		adv_type	number. 0: default alarm info frame 1: Eddystone-uid 2: iBeacon
		channel_type	number. 0: normal adv type 1: trigger-after adv type 2: trigger before and after adv type
		normal_adv	object. valid when channel_type is 0
		adv_interval	number. advertising interval, unit: ms
		tx_power	number. TX power, unit: dBm
		trigger_before_adv	object. valid when channel_type is 2
		adv_interval	number. advertising interval, unit: ms
		tx_power	number. TX power, unit: dBm
		trigger_after_adv	object. valid when channel_type is 1/2
		adv_interval	number. advertising interval, unit: ms
		tx_power	number. TX power, unit: dBm


```
{
  "msg_id" : 3121,
  "device_info" : {
     "mac" : "142b2fe9b19c"
  },
  "data" : {
     "mac" : "fdcbe50c4987",
     "result_code" : 0,
     "adv_param" : [ {
        "channel" : 0,
        "enable" : 1,
        "channel_type" : 0,
        "adv_type" : 0,
```

```
        "normal_adv" : {
          "adv_interval" : 1000,
          "tx_power" : 0
        }
      }, {
        "channel" : 1,
        "enable" : 1,
        "channel_type" : 0,
        "adv_type" : 0,
        "normal_adv" : {
          "adv_interval" : 1000,
          "tx_power" : 0
        }
      }, {
        "channel" : 2,
        "enable" : 0
      }, {
        "channel" : 3,
        "enable" : 0
      } ]
    }
}
```


3.22 Set broadcast parameters (121)

To set broadcast parameters for MK Button device via gateway.

Flag	Msg id offset	Parameter name	Parameter value
Write	122	mac	string. 12 characters (6 bytes of BLE device mac address)
		channel	number. 0~3
		normal_adv	object. normal adv type is optional
		adv_interval	number. 20-10000,unit: ms
		tx_power	number. {-40，-20，-16，-12，-8，-4,0,3,4}
		trigger_before_adv	object. Trigger before adv type is optional
		adv_interval	number. 20-10000,unit: ms
		tx_power	number. {-40，-20，-16，-12，-8，-4,0,3,4}
		trigger_after_adv	object.

			Trigger before adv type is optional
		adv_interval	number. 20-10000,unit: ms
		tx_power	number. {-40，-20，-16，-12，-8，-4,0,3,4}

Setup:


```
{
   "msg_id": 1122,
   "device_info": {
      "mac": "1c6920e45214"
   },
   "data": {
      "mac": "d63bb235cb9c",
         "channel": 0,
         "trigger_after_adv": {
            "adv_interval": 1000,
            "tx_power": 0
         }
      },
          {
         "channel": 1,
         "normal_adv": {
            "adv_interval": 1000,
            "tx_power": 0
         }
      },
          {
         "channel": 2,
         "trigger_before_adv": {
            "adv_interval": 1000,
            "tx_power": -40
         },
         "trigger_after_adv": {
            "adv_interval": 1000,
            "tx_power": 0
         }
      },
   }
```


3.23 Report broadcast parameters set result (123)

When the gateway gets reply from the MK button device for the Bluetooth broadcast parameters set command, it will report the result to cloud.

Flag	Msg id offset	Parameter name	Parameter value
Notify	123	result_code	number. 0-3 0: command success 1: the BLE device is not connected by gateway 2: command is not supported 3: command timeout
		mac	string. 12 characters (6 bytes of BLE device mac address)


```
{
    "msg_id": 3123,
    "device_info": {
        "mac": "4c11ae8be624"
    },
    "data": {
        "mac": "d9a5fa6f3882",
        "result_code": 0
          }
}
```


4. Connect and Manage BXP-B-CR device

Send command from cloud to the gateway, to make the gateway connect and communicate with the MK Button device.

The MK button devices has two firmware branches: BXP-B-D and BXP-B-CR. This chapter describes the remote commands for BXP-B-CR devices.

4.1 Connect MOKO Button device (150)

Send command from cloud to the gateway, to make the gateway connect with the MK Button device.

Flag	Msg id offset	Parameter name	Parameter value
Write	150	mac	string. 12 characters (6 bytes of BLE device mac address)
		passwd	string. 0-16 characters (0-8 bytes of password).

Setup:


```
{
    "msg_id": 1150,
```

    "device_info": {
        "mac": "4c11ae8be624"
    },
    "data": {
        "mac": "D9A5FA6F3882",
        "passwd": "Moko4321"
    }
}

Reply:

{
    "msg_id": 1150,
    "device_info": {
        "mac": "4c11ae8be624"
    },
    "result_code": 0,
    "result_msg": "success"
}

4.2 Report connection result (151)

After the gateway finish the connection, it will actively report the connection result to cloud. If connecting with MK button device successfully, the gateway will also inquire the device information from the button device and transfer to cloud.

Flag	Msg id offset	Parameter name	Parameter value
Write	151	mac	string. 12 characters (6 bytes of BLE device mac address)
		result_code	number. 0-8 0: Connect succeed 1: Gateway BLE scan disabled 2: Exceed the connectable number (the gateway is connected with an another BLE device) 3: BLE device connected (the BLE device has been connected with gateway) 4: BLE device out of range 5: BLE device un-connectable 6: Connect fail 7: BLE device type error(connection type error) 8: Password error

		result_msg	string. The description for the result code
		product_model	string. Valid when result code is 0
		company_name	string. Valid when result code is 0
		hardware_version	string. Valid when result code is 0
		software_version	string. Valid when result code is 0
		firmware_version	string. Valid when result code is 0
		battery_v	number. Battery voltage, valid when result code is 0
		battery_level	**number.** **Battery percentage, valid when result code is 0** **Added in gateway V2.X.X firmware.**
		single_alarm_num	number. Single press alarm number, valid when result code is 0
		double_alarm_num	number. Double press alarm number, valid when result code is 0
		long_alarm_num	number. Long press alarm number, valid when result code is 0
		sensor_status	number. Valid when result code is 0. ➢ Bit0: 3-axis sensor equipped status. 0- not equipped, 1- equipped. ➢ Bit1: temperature&humidity sensor equipped status. 0- not equipped, 1- equipped. ➢ Bit2: light sensor equipped status. 0- not equipped, 1- equipped.

```
{
	"msg_id":  3151,
```

```
    "device_info": {
        "mac": "4c11ae8be624"
    },
    "data": {
        "mac": "d9a5fa6f3882",
        "result_code": 0,
        "result_msg": "Connect succeed",
        "product_model": "BUTTON",
        "company_name": "MOKO TECHNOLOGY LTD.",
        "hardware_version": "MKBNseries",
        "software_version": "BXP-B-CR",
        "firmware_version": "V1.0.1",
        "battery_v": 3000,
         "battery_level" : 80,
        "single_alarm_num": 18,
        "double_alarm_num": 0,
        "long_alarm_num": 0,
        "sensor_status": 1
  }
}
```


4.3 Get BLE device information (152)

Send command from cloud to the gateway, to make the gateway inquire the device information of the MK Button device.

Flag	Msg id offset	Parameter name	Parameter value
Write	152	mac	string. 12 characters (6 bytes of BLE device mac address)

Setup:


```
{
    "msg_id": 1152,
    "device_info": {
        "mac": "4c11ae8be624"
    },
    "data": {
        "mac": "D9A5FA6F3882"
    }
}
```


Reply:


```
{
```

```
    "msg_id": 1152,
    "device_info": {
        "mac": "4c11ae8be624"
    },
    "result_code": 0,
    "result_msg": "success"
}
```


4.4 Report BLE device information (153)

After the gateway gets device information from the MK-Button device, it will transfer to cloud.

Flag	Msg id offset	Parameter name	Parameter value
Notify	153	mac	string. 12 characters (6 bytes of BLE device mac address)
		result_code	number. 0-2 0: command success 1: the BLE device is not connected by gateway 2: command is not supported
		product_model	string. Valid when the result code is 0
		company_name	string. Valid when the result code is 0
		hardware_version	string. Valid when the result code is 0
		software_version	string. Valid when the result code is 0
		firmware_version	string. Valid when the result code is 0
		sensor_status	number. Valid when result code is 0. ➢ Bit0: 3-axis sensor equipped status. 0- not equipped, 1- equipped. ➢ Bit1: temperature&humidity sensor equipped status. 0- not equipped, 1- equipped. ➢ Bit2: light sensor equipped status. 0- not equipped, 1- equipped.

{
    "msg_id": 3153,
    "device_info": {
        "mac": "4c11ae8be624"
    },
    "data": {
        "mac": "d9a5fa6f3882",
        "result_code": 0,
        "result_msg": "Connect succeed",
        "product_model": "BUTTON",
        "company_name": "MOKO TECHNOLOGY LTD.",
        "hardware_version": "MKBNseries",
        "software_version": "BXP-B-CR",
        "firmware_version": "V2.0.8",
        "sensor_status": 1
    }
}

4.5 Get BLE device status (154)

To inquire the alarm status and battery status of the MK Button device.

Flag	Msg id offset	Parameter name	Parameter value
Write	154	mac	string. 12 characters (6 bytes of BLE device mac address)

Setup:

{
    "msg_id": 1154,
    "device_info": {
        "mac": "4c11ae8be624"
    },
    "data": {
        "mac": "D9A5FA6F3882"
    }
}

Reply:

{
    "msg_id": 1154,
    "device_info": {
        "mac": "4c11ae8be624"
    },
    "result_code": 0,

    "result_msg": "success"
}

4.6 Report BLE device status (155)

After the gateway gets device status from the MK Button device, it will transfer the reply to cloud.

Flag	Msg id offset	Parameter name	Parameter value
Notify	155	mac	string. 12 characters (6 bytes of BLE device mac address)
		result_code	number. 0-3 0: command success 1: the BLE device is not connected by gateway 2: command is not supported 3: command timeout
		battery_v	number. Battery voltage, valid when the result code is 0
		battery_level	**number.** **Battery percentage, valid when result code is 0** **Added in gateway V2.X.X firmware.**
		single_alarm_num	number. single press alarm number. Valid when the result code is 0
		double_alarm_num	number. double press alarm number. Valid when the result code is 0
		long_alarm_num	number. long press alarm number. Valid when the result code is 0

{
    "msg_id": 3105,
    "device_info": {
        "mac": "4c11ae8be624"
    },
    "data": {
        "mac": "d9a5fa6f3882",
        "result_code": 0,
        "battery_v": 3000,

"battery_level" : 90,
"single_alarm_num": 18,
"double_alarm_num": 0,
"long_alarm_num": 0
}
}

4.7 Dismiss alarm status (156)

Send this command to the gateway to dismiss the alarm status of the MK- Button device.

Flag	Msg id offset	Parameter name	Parameter value
Write	156	mac	string. 12 characters (6 bytes of BLE device mac address)

Setup:

```
{
    "msg_id": 1156,
    "device_info": {
        "mac": "4c11ae8be624"
    },
    "data": {
        "mac": "D9A5FA6F3882"
    }
}
```

Reply:

```
{
    "msg_id": 1156,
    "device_info": {
        "mac": "4c11ae8be624"
    },
    "result_code": 0,
    "result_msg": "success"
}
```

4.8 Report alarm dismiss result (157)

After sending the dismiss alarm command, if the gateway gets reply from the MK-Button device, it will transfer the reply to cloud.

Flag	Msg id offset	Parameter name	Parameter value

Notify	157	mac	string. 12 characters (6 bytes of BLE device mac address)
		result_code	number. 0-3 0: command success 1: the BLE device is not connected by gateway 2: command is not supported 3: command timeout


```
{
	"msg_id": 3157,
	"device_info": {
		"mac": "4c11ae8be624"
	},
	"data": {
		"mac": "d9a5fa6f3882",
		"result_code": 0
	}
}
```


4.9 Setup LED (158)

To setup the LED of the MK Button device.

Flag	Msg id offset	Parameter name	Parameter value
Write	158	mac	string. 12 characters (6 bytes of Button mac address)
		flash_time	LED flash duration. 1-6000, unit: 100 ms
		flash_interval	LED flash interval. 0-100, unit: 100 ms

Setup:


```
{
	"msg_id": 1158,
	"device_info": {
		"mac": "4c11ae8be624"
	},
	"data": {
		"mac": "D9A5FA6F3882",
		"flash_time":100,
		"flash_interval":10
	}
```

```
}
```

Reply:

```
{
    "msg_id": 1158,
    "device_info": {
        "mac": "4c11ae8be624"
    },
    "result_code": 0,
    "result_msg": "success"
}
```

4.10 Report LED setup result (159)

When the gateway gets reply from the MK Button Beacon for the setup LED command, it will report the result to cloud.

Flag	Msg id offset	Parameter name	Parameter value
Notify	159	mac	string. 12 characters (6 bytes of Button mac address)
		result_code	number. 0-3 0: command success 1: the BLE device is not connected by gateway 2: command is not supported 3: command timeout

```
{
    "msg_id": 3159,
    "device_info": {
        "mac": "4c11ae8be624"
    },
    "data": {
        "mac": "d9a5fa6f3882",
        "result_code": 0
    }
}
```

4.11 Setup Buzzer (160)

To set the Buzzer of the MK Button device.

Flag	Msg id offset	Parameter name	Parameter value

Write	160	mac	string. 12 characters (6 bytes of Button mac address)
		ring_time	Buzzer ring duration. 1-6000, unit: 100ms
		ring_interval	Buzzer ring interval. 0-100, unit: 100 ms

Setup:


```
{
	"msg_id": 1160,
	"device_info": {
		"mac": "4c11ae8be624"
	},
	"data": {
		"mac": "D9A5FA6F3882",
		"ring_time":100,
		"ring_interval":10
	}
}
```


Reply:


```
{
	"msg_id": 1160,
	"device_info": {
		"mac": "4c11ae8be624"
	},
	"result_code": 0,
	"result_msg": "success"
}
```


4.12 Report Buzzer setup result (161)

When the gateway gets reply from the MK Button device for the setup buzzer command, it will report the result to cloud.

Flag	Msg id offset	Parameter name	Parameter value
Notify	161	mac	string. 12 characters (6 bytes of Button mac address)
		result_code	number. 0-3 0: command success 1: the BLE device is not connected by gateway 2: command is not supported 3: command timeout

```
{
    "msg_id": 3161,
    "device_info": {
        "mac": "4c11ae8be624"
    },
    "data": {
        "mac": "d9a5fa6f3882",
        "result_code": 0
    }
}
```

4.13 Delete alarm trigger records(162)

Send command from cloud to gateway, to make the gateway delete the alarm trigger record for the MK button device.

Flag	Msg id offset	Parameter name	Parameter value
Write	162	mac	string. 12 characters (6 bytes of BLE device mac address)
		type	number. 0-2 0: delete single press records 1: delete double click records 2: delete long press records

Setup:

```
{
    "msg_id": 1162,
    "device_info": {
        "mac": "4c11ae8be624"
    },
    "data": {
        "mac": "d9a5fa6f3882",
        "type": 0
    }
}
```

Reply:

```
{
    "msg_id": 1162,
    "device_info": {
        "mac": "4c11ae8be624"
    },
    "result_code": 0,
```

```
    "result_msg": "success"
}
```

4.14 Report alarm trigger records delete result (163)

When the gateway gets a reply from the MK Button device for the deleting trigger records command, it will report the result to cloud.

Flag	Msg id offset	Parameter name	Parameter value
Notify	163	mac	string. 12 characters (6 bytes of BLE device mac address)
		type	number. 0~2 0: delete single press record 1: delete double click record 2: delete long press record
		result_code	number. 0-3 0: command success 1: the Button is not connected by gateway 2: command is not supported 3: command timeout

```
{
    "msg_id": 3163,
    "device_info": {
        "mac": "4c11ae8be624"
    },
    "data": {
        "mac": "d9a5fa6f3882",
        "type": 0
        "result_code": 0
    }
}
```

4.15 Set 3-axis data notify switch (164)

To setup the MK Button to enable/ disable 3-axis data notify switch.

Flag	Msg id offset	Parameter name	Parameter value
Write	164	switch_value	number. 0-1 0: disable

			1: enable
		mac	The mac of BLE device connected by gateway

Setup:


```
{
	"msg_id": 1164,
	"device_info": {
		"mac": "4c11ae8be624"
	}
	"data": {
		"mac": "d9a5fa6f3882",
		"switch_value": 0
	}

}
```


Reply:


```
{
	"msg_id": 1164,
	"device_info": {
		"mac": "4c11ae8be624"
	},
	"result_code": 0,
	"result_msg": "success"
}
```


4.16 Report 3-axis data notify switch set result (165)

When the gateway gets a reply from the MK button for the 3-axis notify switch setup command, it will report the result to cloud.

Flag	Msg id offset	Parameter name	Parameter value
Notify	165	mac	string. 12 characters BLE device MAC address
		result_code	number. 0-3 0: command success 1: the BLE device is not connected with gateway 2: command is not supported 3: command timeout


```
{
	"msg_id": 3165,
```

```
    "device_info": {
        "mac": "4c11ae8be624"
    },
    "data": {
        "mac": "D9A5FA6F3882",
        "result_code": 0
    }
}
```

4.17 Report 3-axis data (166)

When the gateway enables the 3-axis data notify switch, the gateway will receive real time 3-axis data from the MK Button device, then gateway will transfer 3-axis data to cloud.

Flag	Msg id offset	Parameter name	Parameter value
Notify	166	mac	string. 12 characters BLE device MAC address
		x_axis_data	number, unit: mg
		y_axis_data	number, unit: mg
		z_axis_data	number, unit: mg

```
{
    "msg_id": 3166,
    "device_info": {
        "mac": "4c11ae8be624"
    },
    "data": {
        "mac": "d9a5fa6f3882",
        "x_axis_data": 20,
        "y_axis_data": 30,
        "z_axis_data": 20
    }
}
```

4.18 Set device power off (167)

To power off the beacon via gateway.

Flag	Msg id offset	Parameter name	Parameter value
Write	167	mac	string. 12 characters (6 bytes of BLE device mac address)

Setup:


```
{
    "msg_id": 1167,
    "device_info": {
        "mac": "4c11ae8be624"
    },
    "data": {
        "mac": "d9a5fa6f3882"
         }
}
```


4.19 Report set device power off result (168)

When the gateway gets a reply from the MK Button device for set power off command, it will transfer the result to cloud.

Flag	Msg id offset	Parameter name	Parameter value
Notify	168	result_code	number. 0-3 0: command success 1: the Button is not connected by gateway 2: command is not supported 3: command timeout
		mac	string. 12 characters (6 bytes of BLE device mac address)


```
{
    "msg_id": 3168,
    "device_info": {
        "mac": "4c11ae8be624"
    },
    "data": {
        "mac": "d9a5fa6f3882",
        "result_code": 0
         }
}
```


4.20 Setup motor (169)

To set the motor of the MK Button device.

Flag	Msg id offset	Parameter name	Parameter value
Write	169	mac	string. 12 characters (6 bytes of BLE device mac address)
		shake_time	Vibration duration. 1-6000, unit: 100ms
		shake_interval	vibration interval. 0-100, unit: 100 ms

Setup:


```
{
	"msg_id": 1169,
	"device_info": {
		"mac": "4c11ae8be624"
	},
	"data": {
		"mac": "D9A5FA6F3882",
		"ring_time":100,
		"ring_interval":10
	}
}
```


Reply:


```
{
	"msg_id": 1169,
	"device_info": {
		"mac": "4c11ae8be624"
	},
	"result_code": 0,
	"result_msg": "success"
}
```


4.21 Report motor setup result (170)

When the gateway gets reply from the MK Button device for the set motor command, it will report the result to cloud.

Flag	Msg id offset	Parameter name	Parameter value
Notify	170	mac	string. 12 characters (6 bytes of BLE device mac address)
		result_code	number. 0-3 0: command success 1: the BLE device is not connected by gateway

			2: command is not supported 3: command timeout

```
{
	"msg_id": 3170,
	"device_info": {
		"mac": "4c11ae8be624"
	},
	"data": {
		"mac": "d9a5fa6f3882",
		"result_code": 0
	}
}
```

4.22 Set trigger record notify record (171)

To set the trigger data notify switch for the MK Button device.

Flag	Msg id offset	Parameter name	Parameter value
Write	171	mac	The mac of BLE devices connected by gateway
		switch_value	number. 0: disable 1: enable
		type	number. 0: single press trigger record 1: double press trigger record 2: long press trigger record

Setup:

```
{
	"msg_id": 1171,
	"device_info": {
		"mac": "4c11ae8be624"
	}
	"data": {
		"mac": "d9a5fa6f3882",
		"switch_value": 0,
		"type": 0
	}

}
```

Reply:

```
{
    "msg_id": 1171,
    "device_info": {
        "mac": "4c11ae8be624"
    },
    "result_code": 0,
    "result_msg": "success"
}
```


4.23 Report trigger record notify switch set result (172)

When the gateway gets reply from the MK Button device for the trigger record notify switch setup command, it will report the result to cloud.

Flag	Msg id offset	Parameter name	Parameter value
Notify	172	mac	string. 12 characters BLE device MAC address
		result_code	number. 0-3 0: command success 1: the BLE device is not connected by gateway 2: command is not supported 3: command timeout


```
{
    "msg_id": 3172,
    "device_info": {
        "mac": "4c11ae8be624"
    },
    "data": {
        "mac": "D9A5FA6F3882",
        "result_code": 0
    }
}
```


4.24 Report trigger record data (173)

When the gateway enable trigger record notify switch, it will receive the trigger record data and report to cloud.

Flag	Msg id offset	Parameter name	Parameter value

Notify	173	mac	string. 12 characters BLE device MAC address
		type	number 0: single press 1: double press 2: long press
		timestamp	number, unix timestamp


```
{
	"msg_id": 3173,
	"device_info": {
		"mac": "4c11ae8be624"
	},
	"data": {
		"mac": "d9a5fa6f3882",
		"type": 0,
		"timestamp": 1747045678708
	}
}
```


4.25 Get broadcast parameters (174)

To inquire the Bluetooth broadcast parameters of MK button device.

Flag	Msg id offset	Parameter name	Parameter value
Write	174	mac	string. 12 characters (6 bytes of BLE device mac address)

Setup:


```
{
	"msg_id": 1174,
	"device_info": {
		"mac": "4c11ae8be624"
	},
	"data": {
		"mac": "d9a5fa6f3882"
	}
}
```


4.26 Report broadcast parameters (175)

When the gateway gets reply from the MK Button device for the getting Bluetooth broadcast parameters command, it will transfer the result to cloud.

Flag	Msg id offset	Parameter name	Parameter value
Notify	121	mac	string. 12 characters (6 bytes of BLE device mac address)
		result_code	number. 0-3 0: command success 1: the BLE device is not connected by gateway 2: command is not supported 3: command timeout
		adv_param	Array. valid when result code is 0. The BLE device has four adv channels, therefore it has at most 4 groups of adv parameters.
		channel	number. 0-3
		enable	number. 0: disable 1: enable
		channel_type	number. 0: normal adv type 1: trigger-after adv type 2: trigger before and after adv type
		normal_adv	object. valid when channel_type is 0
		adv_interval	number. 20-10000, unit: ms
		tx_power	number. {-40,-20,-16,-12,-8,-4,0,3,4}
		trigger_before_adv	object. valid when channel_type is 2
		adv_interval	number. advertising interval, unit: ms
		tx_power	number. TX power, unit: dBm
		trigger_after_adv	object. valid when channel_type is 1/2
		adv_interval	number. 20-10000, unit: ms
		tx_power	number. {-40,-20,-16,-12,-8,-4,0,3,4}

```
{
   "msg_id":  3175,
   "device_info":  {
      "mac":  "1c6920e45214"
   },
   "data":  {
      "mac":  "d63bb235cb9c",
      "result_code":  0,
      "adv_param":  [{
         "channel":  0,
         "enable":  1,
         "channel_type":  1,
         "trigger_after_adv":  {
            "adv_interval":  1000,
            "tx_power":  0
         }
      },
          {
         "channel":  1,
         "enable":  1,
         "channel_type":  0,
         "normal_adv":  {
            "adv_interval":  1000,
            "tx_power":  0
         }
      },
          {
         "channel":  2,
         "enable":  1,
         "channel_type":  2,
         "trigger_before_adv":  {
            "adv_interval":  1000,
            "tx_power":  -40
         },
         "trigger_after_adv":  {
            "adv_interval":  1000,
            "tx_power":  0
         }
      },  {
         "channel":  3,
         "enable":  0
      }  ]
   }
}
```

4.27 Set broadcast parameters (176)

To set broadcast parameters for MK Button device via gateway.

Flag	Msg id offset	Parameter name	Parameter value
Write	176	mac	string. 12 characters (6 bytes of BLE device mac address)
		channel	number. 0-3
		normal_adv	object. normal adv type is optional
		adv_interval	number. 20-10000, unit: ms
		tx_power	number. {-40,-20,-16,-12,-8,-4,0,3,4}
		trigger_before_adv	object. Trigger before adv type is optional
		adv_interval	number. 20-10000, unit: ms
		tx_power	number. {-40,-20,-16,-12,-8,-4,0,3,4}
		trigger_after_adv	object. Trigger before adv type is optional
		adv_interval	number. 20-10000, unit: ms
		tx_power	number. {-40,-20,-16,-12,-8,-4,0,3,4}

Setup:


```
{
  "msg_id": 1176,
  "device_info": {
    "mac": "1c6920e45214"
  },
  "data": {
    "mac": "d63bb235cb9c",
      "channel": 0,
      "trigger_after_adv": {
        "adv_interval": 1000,
        "tx_power": 0
      }
    },
```

```
    {
    "channel": 1,
    "normal_adv": {
        "adv_interval": 1000,
        "tx_power": 0
    }
 },
    {
    "channel": 2,
    "trigger_before_adv": {
        "adv_interval": 1000,
        "tx_power": -40
    },
    "trigger_after_adv": {
        "adv_interval": 1000,
        "tx_power": 0
    }
 },
}
```

4.28 Report broadcast parameters set result (177)

When the gateway gets reply from the MK button device for the Bluetooth broadcast parameters set command, it will report the result to cloud.

Flag	Msg id offset	Parameter name	Parameter value
Notify	123	result_code	number. 0-3 0: command success 1: the BLE device is not connected by gateway 2: command is not supported 3: command timeout
		mac	string. 12 characters (6 bytes of BLE device mac address)


```
{
    "msg_id": 3177,
    "device_info": {
        "mac": "4c11ae8be624"
    },
    "data": {
        "mac": "d9a5fa6f3882",
        "result_code": 0
        }
```

```
}
```

5. Connect and Manage BXP-C device

Send command from cloud to the gateway, to make the gateway connect and communicate with the BXP-C device.

5.1 Connect MOKO BXP-C device (350)

To make the gateway connect to the MOKO BXP-C device.

Flag	Msg id offset	Parameter name	Parameter value
Write	350	mac	string. 12 characters (6 bytes of BLE device mac address)
		passwd	string. 0-16 characters (0-8 bytes of password).

Setup:

```
{
	"msg_id": 1350,
	"device_info": {
		"mac": "4c11ae8be624"
	},
	"data": {
		"mac": "D9A5FA6F3882",
		"passwd": "Moko4321"
	}
}
```

Reply:

```
{
	"msg_id": 1350,
	"device_info": {
		"mac": "4c11ae8be624"
	},
	"result_code": 0,
	"result_msg": "success"
}
```

5.2 Report connection result (351)

After the gateway connects the BXP-C device, the gateway will actively report the connection result. If connect successfully, the gateway will also inquire device information and report to cloud together.

Flag	Msg id offset	Parameter name	Parameter value
Notify	351	mac	string. 12 characters (6 bytes of BLE device mac address)
		result_code	number. 0-8 0: Connect succeed 1: Gateway BLE scan disabled 2: Exceed the connectable number (the gateway is connected with an another BLE device) 3: BLE device connected (the BLE device has been connected with gateway) 4: BLE device out of range 5: BLE device un-connectable 6: Connect fail 7: BLE device type error(connection type error) 8: Password error
		result_msg	string. The description for the result code
		product_model	string. Valid when result code is 0
		company_name	string. Valid when result code is 0
		hardware_version	string. Valid when result code is 0
		software_version	string. Valid when result code is 0
		firmware_version	string. Valid when result code is 0
		battery_level	number. Valid when result code is 0
		sensor_status	number. Valid when result code is 0. ➢ Bit0: 3-axis sensor equipped status. 0- not

			equipped, 1- equipped. ➢ Bit1: temperature&humidity sensor equipped status. 0- not equipped, 1- equipped. ➢ Bit2: light sensor equipped status. 0- not equipped, 1- equipped.


```
{
	"msg_id": 3351,
	"device_info": {
		"mac": "4c11ae8be624"
	},
	"data": {
		"mac": "d9a5fa6f3882",
		"result_code": 0,
		"result_msg": "Connect succeed",
		"product_model": "BeaconX Pro",
		"company_name": "MOKO TECHNOLOGY LTD.",
		"hardware_version": "MKBNseries",
		"software_version": "BXP-C",
		"firmware_version": "V1.0.1",
		"battery_level": 3000,
		"sensor_status": 1
	}
}
```


5.3 Get device information (352)

To inquire the device information of the BXP-C device.

Flag	Msg id offset	Parameter name	Parameter value
Write	352	mac	string. 12 characters (6 bytes of BLE device mac address)

Setup:


```
{
	"msg_id": 1352,
	"device_info": {
		"mac": "4c11ae8be624"
	},
	"data": {
```

```
        "mac":  "D9A5FA6F3882"
    }
}
```

Reply:

```
{
    "msg_id":  1352,
    "device_info":  {
        "mac":  "4c11ae8be624"
    },
    "result_code":  0,
    "result_msg":  "success"
}
```

5.4 Report device information (353)

After the gateway get device information from the BXP-C device, it will report to cloud.

Flag	Msg id offset	Parameter name	Parameter value
Notify	353	mac	string. 12 characters (6 bytes of Button mac address)
		result_code	number. 0-2 0: success 1: the Button is not connected by gateway 2: command is not supported
		product_model	string. Valid when the result code is 0
		company_name	string. Valid when the result code is 0
		hardware_version	string. Valid when the result code is 0
		software_version	string. Valid when the result code is 0
		firmware_version	string. Valid when the result code is 0
		sensor_status	number. Valid when result code is 0. ➢ Bit0: 3-axis sensor equipped status. 0- not equipped, 1- equipped. ➢ Bit1: temperature&humidity sensor equipped status. 0- not

			equipped, 1- equipped. ➢ Bit2: light sensor equipped status. 0- not equipped, 1- equipped.


```
{
    "msg_id": 3353,
    "device_info": {
        "mac": "4c11ae8be624"
    },
    "data": {
        "mac": "d9a5fa6f3882",
        "result_code": 0,
        "result_msg": "Connect succeed",
        "product_model": "BUTTON",
        "company_name": "MOKO TECHNOLOGY LTD.",
        "hardware_version": "MKBNseries",
        "software_version": "BXP-C",
        "firmware_version": "V1.0.1",
        "sensor_status": 1
    }
}
```


5.5 Get battery level (354)

To get battery status of the BXP-C device via gateway.

Flag	Msg id offset	Parameter name	Parameter value
Write	354	mac	string. 12 characters (6 bytes of BLE device mac address)

Setup:


```
{
    "msg_id": 1354,
    "device_info": {
        "mac": "4c11ae8be624"
    },
    "data": {
        "mac": "D9A5FA6F3882"
    }
}
```


Reply:


```
{
```

```
    "msg_id": 1354,
    "device_info": {
        "mac": "4c11ae8be624"
    },
    "result_code": 0,
    "result_msg": "success"
}
```

5.6 Report battery level (355)

After the gateway get battery level from the BXP-C device, it will report to cloud.

Flag	Msg id offset	Parameter name	Parameter value
Notify	355	mac	string. 12 characters (6 bytes of Button mac address)
		result_code	number. 0-3 0: command success 1: the BLE device is not connected with gateway 2: command is not supported 3: command timeout
		battery_level	number. Valid when the result code is 0

```
{
    "msg_id": 3355,
    "device_info": {
        "mac": "4c11ae8be624"
    },
    "data": {
        "mac": "d9a5fa6f3882",
        "result_code": 0,
        "battery_level": 30
    }
}
```

5.7 Set real-time T&H data notify switch (356)

To enable/ disable the real-time temperature &humidity notify switch.

Flag	Msg id offset	Parameter name	Parameter value

Write	356	switch_value	number. 0: disable 1: enable
		mac	string. 12 characters (6 bytes of BLE device mac address)

Setup:


```
{
    "msg_id": 1356,
    "device_info": {
        "mac": "4c11ae8be624"
    },
    "data": {
        "mac": "D9A5FA6F3882",
        "switch_value": 1
    }
}
```


Device reply:


```
{
    "msg_id": 1356,
    "device_info": {
        "mac": "4c11ae8be624"
    },
    "result_code": 0,
    "result_msg": "success"
}
```


5.8 Report real-time T&H data notify switch set result (357)

After the gateway gets reply for the notify switch set command, it will report the result to cloud.

Flag	Msg id offset	Parameter name	Parameter value
Notify	357	mac	string. 12 characters (6 bytes of Button mac address)
		result_code	number. 0-2 0: success 1: the Button is not connected by gateway 2: command is not supported 3: command time out

```
{
    "msg_id": 3357,
    "device_info": {
        "mac": "4c11ae8be624"
    },
    "data": {
        "mac": "d9a5fa6f3882",
        "result_code": 0
    }
}
```

5.9 Notify real-time T&H data (358)

When the gateway enables the real time temperature and humidity data notify switch, the gateway will receive real time temperature and humidity data from the device, then gateway will transfer to cloud.

Flag	Msg id offset	Parameter name	Parameter value
Notify	358	mac	string. 12 characters (6 bytes of BLE device mac address)
		temperature	number, unit 0.1° C
		humidity	number. unit 0.1%

```
{
    "msg_id": 3358,
    "device_info": {
        "mac": "4c11ae8be624"
    },
    "data": {
        "mac": "d9a5fa6f3882",
        "temperature": 301,
        "humidity": 281
    }
}
```

5.10 Set 3-axis data notify switch (359)

To setup the BXP-C device to enable/ disable 3-axis data notify switch.

Flag	Msg id offset	Parameter name	Parameter value

Write	359	switch_value	number. 0-1 0: disable 1: enable
		mac	The mac of BLE device connected by gateway

Setup:


```
{
    "msg_id": 1359,
    "device_info": {
        "mac": "4c11ae8be624"
    }
   "data": {
        "mac": "d9a5fa6f3882",
        "switch_value": 0
    }

}
```


Reply:


```
{
    "msg_id": 1359,
    "device_info": {
        "mac": "4c11ae8be624"
    },
    "result_code": 0,
    "result_msg": "success"
}
```


5.11 Report 3-axis data notify switch set result (360)

When the gateway gets a reply from the BLE device for the 3-axis notify switch setup command, it will report the result to cloud.

Flag	Msg id offset	Parameter name	Parameter value
Notify	360	mac	string. 12 characters BLE device MAC address
		result_code	number. 0-3 0: command success 1: the BLE device is not connected with gateway 2: command is not supported 3: command timeout

```
{
    "msg_id": 3360,
    "device_info": {
        "mac": "4c11ae8be624"
    },
    "data": {
        "mac": "D9A5FA6F3882",
        "result_code": 0
    }
}
```

5.12 Report 3-axis data (361)

When the gateway enables the 3-axis data notify switch, the gateway will receive real time 3-axis data from the BXP-C device, then gateway will transfer to cloud.

Flag	Msg id offset	Parameter name	Parameter value
Notify	361	mac	string. 12 characters BLE device MAC address
		x_axis_data	number
		y_axis_data	number
		z_axis_data	number.

```
{
    "msg_id": 3361,
    "device_info": {
        "mac": "4c11ae8be624"
    },
    "data": {
        "mac": "d9a5fa6f3882",
        "x_axis_data": 200,
        "y_axis_data": 305,
        "z_axis_data": 20
    }
}
```

5.13 Set history T&H data notify switch(362)

To set the history temperature &humidity notify switch for BXP-C device via gateway.

Flag	Msg id offset	Parameter name	Parameter value
Write	362	switch_value	number.

			0: disable 1: enable
		mac	string. 12 characters (6 bytes of BLE device mac address)

Setup:

```
{
      "msg_id": 1362,
      "device_info": {
            "mac": "4c11ae8be624"
      },
      "data": {
            "mac": "d9a5fa6f3882",
            "switch_value": 1
               }
}
```

5.14 Report history T&H data notify switch set result (363)

After the gateway gets reply for the notify switch set command, it will report the result to cloud.

Flag	Msg id offset	Parameter name	Parameter value
Notify	363	result_code	number. 0-3 0: success 1: the Button is not connected by gateway 2: command is not supported 3: command timeout
		mac	string. 12 characters (6 bytes of BLE mac address)

```
{
      "msg_id": 3363,
      "device_info": {
            "mac": "4c11ae8be624"
      },
      "data": {
            "mac": "d9a5fa6f3882",
            "result_code": 0
               }
```

```
}
```

5.15 Report history T&H data (364)

When the gateway enables the history temperature and humidity notify switch, the gateway will receive history temperature and humidity data from the device, then gateway will transfer to cloud.

Flag	Msg id offset	Parameter name	Parameter value
Notify	364	timestamp	number. unit: second
		temperature	number. unit: 0.1℃
		humidity	number. unit: 0.1%
		mac	string. 12 characters (6 bytes of BLE device mac address)


```
{
    "msg_id": 3364,
    "device_info": {
        "mac": "4c11ae8be624"
    },
    "data": {
        "mac": "d9a5fa6f3882",
        "timestamp": 1747103553,
        "temperature": 256,
        "humidity": 512
    }
}
```


5.16 Delete history T&H data (365)

To clear history temperature &humidity data from BXP-C device.

Flag	Msg id offset	Parameter name	Parameter value
Write	365	mac	string. 12 characters (6 bytes of BLE device mac address)

Setup:


```
{
    "msg_id": 1365,
    "device_info": {
        "mac": "4c11ae8be624"
    },
    "data": {
```

```
        "mac": "D9A5FA6F3882"
    }
}
```

Reply:

```
{
    "msg_id": 1365,
    "device_info": {
        "mac": "4c11ae8be624"
    },
    "result_code": 0,
    "result_msg": "success"
}
```


5.17 Report history T&H data delete result (366)

When the gateway gets a reply from the BXP-C device for the delete history temperature and humidity command, it will transfer to cloud.

Flag	Msg id offset	Parameter name	Parameter value
Notify	366	mac	string. 12 characters (6 bytes of BLE device mac address)
		result_code	number. 0-3 0: success 1: the BLE device is not connected by gateway 2: command is not supported 3: command timeout


```
{
    "msg_id": 3366,
    "device_info": {
        "mac": "4c11ae8be624"
    },
    "data": {
        "mac": "d9a5fa6f3882",
        "result_code": 0
    }
}
```

5.18 Set device power off (367)

To power off the BXP-C device via gateway.

Flag	Msg id offset	Parameter name	Parameter value
Write	367	mac	string. 12 characters (6 bytes of BLE device mac address)

Setup:


```
{
      "msg_id":  1367,
      "device_info":  {
            "mac":  "4c11ae8be624"
      },
      "data":  {
            "mac":  "d9a5fa6f3882"
            }
}
```


5.19 Report set device power off result (368)

When the gateway gets a reply from the BXP-C device for control power off command, it will transfer the result to cloud.

Flag	Msg id offset	Parameter name	Parameter value
Notify	368	result_code	number. 0-3 0: command success 1: the BLE device is not connected by gateway 2: command is not supported 3: command timeout
		mac	string. 12 characters (6 bytes of BLE device mac address)


```
{
      "msg_id":  3168,
      "device_info":  {
            "mac":  "4c11ae8be624"
      },
      "data":  {
            "mac":  "d9a5fa6f3882",
            "result_code":  0
            }
```

```
}
```

5.20 Get T&H sensor sampling rate (369)

To get the temperature &humidity sampling rate of the device.

Flag	Msg id offset	Parameter name	Parameter value
Write	369	mac	string. 12 characters BLE device MAC address

Setup:


```
{
	"msg_id":  1369,
	"device_info":  {
		"mac":  "4c11ae8be624"
	},
	"data":  {
		"mac":  "d9a5fa6f3882"
	}
}
```


5.21 Report T&H sensor sampling rate (370)

When the gateway gets a reply from the BXP-C device for the getting temperature &humidity sampling rate command, it will notify the result to cloud.

Flag	Msg id offset	Parameter name	Parameter value
Write	370	result_code	number. 0-3 0: command success 1: the ble device is not connected by gateway 2: command is not supported 3: command timeout
		sampling_rate	number valid when result_code is 0
		mac	string. 12 characters BLE device MAC address

Setup:


```
{
	"msg_id":  3370,
	"device_info":  {
```

"mac": "4c11ae8be624"
},
"data": {
"mac": "d9a5fa6f3882",
"result_code": "0,
"sampling_rate": 10
}
}

5.22 Set T&H sensor sampling rate (371)

To set the temperature &humidity sampling rate for BLE device.

Flag	Msg id offset	Parameter name	Parameter value
Write	371	mac	string. 12 characters BLE device MAC address
		sampling_rate	number. 1-65535,unit: second

Setup:

{
"msg_id": 1371,
"device_info": {
"mac": "4c11ae8be624"
},
"data": {
"mac": "d9a5fa6f3882",
"sampling_rate": 10
}
}

5.23 Report T&H sensor sampling rate set result (372)

When the gateway gets a reply from the BLE Beacon for the temperature &humidity sampling rate set command, it will report the result to cloud.

Flag	Msg id offset	Parameter name	Parameter value
Write	372	result_code	number. 0-3 0: command success 1: the BLE device is not connected by gateway 2: command is not supported 3: command timeout

		mac	string. 12 characters BLE device MAC address

Setup:

```
{
      "msg_id": 3372,
      "device_info": {
            "mac": "4c11ae8be624"
      },
      "data": {
            "mac": "d9a5fa6f3882",
            "result_code": "0
      }
}
```

5.24 Get broadcast parameters (373)

To get the Bluetooth broadcast parameters of BLE device.

Flag	Msg id offset	Parameter name	Parameter value
Write	373	mac	string. 12 characters (6 bytes of BLE device mac address)

Setup:

```
{
      "msg_id": 1373,
      "device_info": {
            "mac": "4c11ae8be624"
      },
      "data": {
            "mac": "d9a5fa6f3882",
               }
}
```

5.25 Report broadcast parameters (374)

When the gateway gets the broadcast parameters from the BLE device, it will report to cloud.

Flag	Msg id offset	Parameter name	Parameter value
Notify	374	mac	string. 12 characters (6 bytes of BLE device mac address)
		result_code	number. 0-3 0: command success

			1: the BLE device is not connected by gateway 2: command is not supported 3: command timeout
		adv_param	Array. The device has 6 adv slots, therefore it will have max 6 groups of advertising parameters. valid when result code is 0
		channel	number. 0~5
		channel_type	number. 0x00: Eddystone-UID 0x10: Eddystone-URL 0x20: Eddystone-TLM 0x30: Eddystone-EID 0x40: BXP-Device info 0x50: iBeacon 0x60: BXP-ACC 0x70: BXP-T&H 0xff: disable
		trigger_type	number. 0: no trigger 1: Temperature trigger 2: humidity trigger 3: double press trigger 4: Triple press trigger 5: Movement trigger 6: Light sensor trigger
		adv_interval	number. unit: second
		tx_power	number. unit: dBm


```
{
  "msg_id": 3374,
  "device_info": {
    "mac": "1c6920e45214"
  },
  "data": {
    "mac": "dd4b9c105f47",
    "result_code": 0,
    "adv_param": [
      "channel": 0,
      "channel_type": 1,
      "trigger_type": 0,
```

```
        "adv_interval":  1000,
        "tx_power":  0
          ]
    }
}
```

5.26 Set broadcast parameters (375)

To set Bluetooth broadcast parameters for BLE device.

Flag	Msg id offset	Parameter name	Parameter value
Write	375	mac	string. 12 characters (6 bytes of Button mac address)
		channel	number. 0~5
		adv_interval	number. 100~10000, unit: ms
		tx_power	number. Options: -40,-20,-16,-12,-8,-4,0,3,4 unit: dBm

Setup:

```
{
     "msg_id":  1375,
     "device_info":  {
          "mac":  "4c11ae8be624"
     },
     "data":  {
          "mac":  "d9a5fa6f3882",
          "channel":  1,
          "adv_interval":  100,
          "tx_power":  0

            }
}
```

5.27 Report broadcast parameters set result (376)

When the gateway gets a reply from the BXP-C Beacon for remote control power off command, it will notify the result to cloud.

Flag	Msg id offset	Parameter name	Parameter value
Notify	376	result_code	number. 0-3 0: command success

			1: the Button is not connected by gateway 2: command is not supported 3: command timeout
		mac	string. 12 characters (6 bytes of BLE mac address)


```
{
      "msg_id":  3376,
      "device_info":  {
            "mac":  "4c11ae8be624"
      },
      "data":  {
            "mac":  "d9a5fa6f3882",
            "result_code":  0
               }
}
```


6. Connect and Manage BXP-D device

Send command from cloud to the gateway, to make the gateway connect and communicate with the BXP-D device.

6.1 Connect MOKO BXP-D device (400)

To make the gateway connect to the MOKO BXP-D device.

Flag	Msg id offset	Parameter name	Parameter value
Write	400	mac	string. 12 characters (6 bytes of BLE device mac address)
		passwd	string. 0-16 characters (0-8 bytes of password).

Setup:


```
{
      "msg_id":  1400,
      "device_info":  {
            "mac":  "4c11ae8be624"
      },
      "data":  {
```

```
            "mac":  "D9A5FA6F3882",
            "passwd":  "Moko4321"
      }
}
```

Device reply:

```
{
      "msg_id":  1400,
      "device_info":  {
            "mac":  "4c11ae8be624"
      },
      "result_code":  0,
      "result_msg":  "success"
}
```


6.2 Report connection result (401)

After the gateway connects the BXP-D device, the gateway will actively report the connection result. If connect successfully, the gateway will also inquire device information and report to cloud together.

Flag	Msg id offset	Parameter name	Parameter value
Notify	401	mac	string. 12 characters (6 bytes of BLE device mac address)
		result_code	number. 0-8 0: Connect succeed 1: Gateway BLE scan disabled 2: Exceed the connectable number (the gateway is connected with an another BLE device) 3: BLE device connected (the BLE device has been connected with gateway) 4: BLE device out of range 5: BLE device un-connectable 6: Connect fail 7: BLE device type error(connection type error) 8: Password error
		result_msg	string. The description for the result code
		product_model	string.

			Valid when result code is 0
		company_name	string. Valid when result code is 0
		hardware_version	string. Valid when result code is 0
		software_version	string. Valid when result code is 0
		firmware_version	string. Valid when result code is 0
		battery_v	number. Battery voltage, unit: mV Valid when result code is 0


```
{
    "msg_id": 3401,
    "device_info": {
        "mac": "4c11ae8be624"
    },
    "data": {
        "mac": "d9a5fa6f3882",
        "result_code": 0,
        "result_msg": "Connect succeed",
        "product_model": "BUTTON",
        "company_name": "MOKO TECHNOLOGY LTD.",
        "hardware_version": "MKBNseries",
        "software_version": "BXP-B-D",
        "firmware_version": "V1.0.1",
        "battery_v": 3000
  }
}
```


6.3 Get device information (402)

To inquire the device information of the BXP-D device.

Flag	Msg id offset	Parameter name	Parameter value
Write	402	mac	string. 12 characters (6 bytes of BLE device mac address)

Setup:


```
{
    "msg_id": 1402,
```

```
    "device_info": {
        "mac": "4c11ae8be624"
    },
    "data": {
        "mac": "D9A5FA6F3882"
    }
}
```

Reply:

```
{
    "msg_id": 1402,
    "device_info": {
        "mac": "4c11ae8be624"
    },
    "result_code": 0,
    "result_msg": "success"
}
```

6.4 Report device information (403)

After the gateway get device information from the BXP-D device, it will report to cloud.

Flag	Msg id offset	Parameter name	Parameter value
Notify	403	mac	string. 12 characters (6 bytes of Button mac address)
		result_code	number. 0-2 0: success 1: the Button is not connected by gateway 2: command is not supported
		product_model	string. Valid when the result code is 0
		company_name	string. Valid when the result code is 0
		hardware_version	string. Valid when the result code is 0
		software_version	string. Valid when the result code is 0
		firmware_version	string. Valid when the result code is 0

```
{
    "msg_id": 3353,
```

```
    "device_info": {
        "mac": "4c11ae8be624"
    },
        "data" : {
        "mac" : "f505ecb011dd",
        "result_code" : 0,
        "product_model" : "BeaconX Pro",
        "company_name" : "MOKO TECHNOLOGY LTD.",
        "hardware_version" : "MKBN Series",
        "software_version" : "nRF52-SDK14.2",
        "firmware_version" : "BXP-D_V3.0.14"
      }
}
```

6.5 Get battery level (404)

To get battery status of the BXP-D device via gateway.

Flag	Msg id offset	Parameter name	Parameter value
Write	404	mac	string. 12 characters (6 bytes of BLE device mac address)

Setup:

```
{
    "msg_id": 1404,
    "device_info": {
        "mac": "4c11ae8be624"
    },
    "data": {
        "mac": "D9A5FA6F3882"
    }
}
```

Device reply:

```
{
    "msg_id": 1404,
    "device_info": {
        "mac": "4c11ae8be624"
    },
    "result_code": 0,
    "result_msg": "success"
}
```

6.6 Report battery level (405)

After the gateway get battery level from the BXP-D device, it will report to cloud.

Flag	Msg id offset	Parameter name	Parameter value
Notify	405	mac	string. 12 characters (6 bytes of Button mac address)
		result_code	number. 0-3 0: command success 1: the BLE device is not connected with gateway 2: command is not supported 3: command timeout
		battery_v	number. Battery voltage Valid when the result code is 0


```
{
      "msg_id":  3405,
      "device_info":  {
            "mac":  "4c11ae8be624"
      },
      "data":  {
            "mac":  "d9a5fa6f3882",
            "result_code":  0,
            "battery_v":  3200
      }
}
```


6.7 Get 3-axis parameters(406)

To get the 3-axis parameters of BLE device.

Flag	Msg id offset	Parameter name	Parameter value
Write	406	mac	The mac of BLE devices connected by gateway

setup:


```
{
      "msg_id":  1406,
      "device_info":  {
            "mac":  "4c11ae8be624"
      }
     "data":  {
```

```
        "mac": "d9a5fa6f3882"
    }

}
```
Reply:
```
{
    "msg_id": 1406,
    "device_info": {
        "mac": "4c11ae8be624"
    },
    "result_code": 0,
    "result_msg": "success"
}
```

6.8 Report 3-axis parameters (407)

When the gateway gets a reply from the BXP-D device for the inquire 3-axis parameters command, it will report the result to cloud.

Flag	Msg id offset	Parameter name	Parameter value
Notify	407	mac	string. 12 characters BLE device MAC address
		result_code	number. 0-3 0: success 1: the Button is not connected by gateway 2: command is not supported 3: command timeout
		full_scale	number 0:2g 1:4g 2:8g 3:16g valid when result_code is 0
		sensitivity	number, unit: 100mg valid when result_code is 0
		sampling_rate	number 0:1hz 1:10hz 2:25hz 3:50hz 4:100hz valid when result_code is 0

```
{
    "msg_id": 3407,
    "device_info": {
        "mac": "4c11ae8be624"
    },
    "data": {
        "mac": "D9A5FA6F3882",
        "result_code": 0,
        "full_scale": 0,
        "sensitivity": 1,
        "sampling_rate": 1
    }
}
```

6.9 Set 3-axis parameters (408)

To set 3-axis parameters for BLE device.

Flag	Msg id offset	Parameter name	Parameter value
Write	408	mac	string. 12 characters BLE device MAC address
		full_scale	number 0:2g 1:4g 2:8g 3:16g valid when result_code is 0
		sensitivity	number, 1-160, unit: 100mg When full scale is 2g, sensitivity is no greater than 20; When full scale is 4g, sensitivity is no greater than 40; When full scale is 8g, sensitivity is no greater than 80; When full scale is 16g, sensitivity is no greater than 160; valid when result_code is 0
		sampling_rate	number 0:1hz 1:10hz 2:25hz 3:50hz

			4:100hz valid when result_code is 0

Setup:


```
{
   "msg_id":  1408,
   "device_info":  {
      "mac":  "1c6920e45214"
   },
   "data":  {
      "mac":  "f0e4b7689d82",
      "full_scale":  4,
      "sensitivity":  2,
      "sampling_rate":  2
   }
}
```


6.10 Report 3-axis parameters set result (409)

After the gateway get reply from the BLE device for the 3-axis parameters set command, it will report to cloud.

Flag	Msg id offset	Parameter name	Parameter value
Notify	409	mac	string. 12 characters (6 bytes of BLE device mac address)
		result_code	number. 0-3 0: success 1: the Button is not connected by gateway 2: command is not supported 3: command timeout


```
{
      "msg_id":  3409,
      "device_info":  {
            "mac":  "4c11ae8be624"
      },
      "data":  {
            "mac":  "d9a5fa6f3882",
            "result_code":  0
      }
}
```

6.11 Get 3-axis sensor resolution (410)

To get 3-axis sensor sample resolution of the BXP-D device.

Flag	Msg id offset	Parameter name	Parameter value
Write	410	mac	string. 12 characters (6 bytes of BLE device mac address)

Setup:


```
{
      "msg_id":  1410,
      "device_info":  {
            "mac":  "4c11ae8be624"
      },
      "data":  {
            "mac":  "d9a5fa6f3882"
               }
}
```


6.12 Report 3-axis sensor resolution (411)

After the gateway get the 3-axis sensor resolution from the BXP-D device, it will report to cloud.

Flag	Msg id offset	Parameter name	Parameter value
Notify	411	result_code	number. 0-3 0: success 1: the BLE device is not connected by gateway 2: command is not supported 3: command timeout
		accuracy	number. 0: 8bit 1: 10bit 2: 12bit
		mac	string. 12 characters (6 bytes of BLE device mac address)


```
{
      "msg_id":  3411,
      "device_info":  {
            "mac":  "4c11ae8be624"
      },
```

```
    "data": {
        "mac": "d9a5fa6f3882",
        "result_code": 0,
        "accuracy": 0
          }
}
```

6.13 Set 3-axis accuracy (412)

To set 3-axis sensor sample resolution for the BXP-D device.

Flag	Msg id offset	Parameter name	Parameter value
Write	412	accuracy	number. 0: 8bit 1: 10bit 2: 12bit
		mac	string. 12 characters (6 bytes of Button mac address)

Setup:

```
{
    "msg_id": 1412,
    "device_info": {
        "mac": "4c11ae8be624"
    },
    "data": {
        "mac": "d9a5fa6f3882",
        "accuracy": 0
          }
}
```

6.14 Report 3-axis resolution set result (413)

After the gateway get reply from the BXP-D device for the 3-axis sensor resolution setup command, it will transfer to cloud.

Flag	Msg id offset	Parameter name	Parameter value
Notify	413	mac	string. 12 characters (6 bytes of Button mac address)
		result_code	number. 0-3 0: success 1: the Button is not connected by

			gateway 2: command is not supported 3: command timeout


```
{
	"msg_id": 3413,
	"device_info": {
		"mac": "4c11ae8be624"
	},
	"data": {
		"mac": "D9A5FA6F3882",
		"result_code": 0
	}
}
```


6.15 Set 3-axis data notify switch(414)

To setup the BXP-D device to enable/ disable 3-axis data notify switch.

Flag	Msg id offset	Parameter name	Parameter value
Write	414	mac	string. 12 characters (6 bytes of BLE device mac address)
		switch_value	number. 0: disable 1: enable

Setup:


```
{
	"msg_id": 1414,
	"device_info": {
		"mac": "4c11ae8be624"
	},
	"data": {
		"mac": "d9a5fa6f3882",
		"switch_value": 1
	}
}
```


6.16 Report 3-axis data notify switch set result (415)

When the gateway gets a reply from the BXP-D device for the 3-axis data notify switch setup command, it will report the result to cloud.

Flag	Msg id offset	Parameter name	Parameter value
Notify	415	result_code	number. 0-3 0: success 1: the Button is not connected by gateway 2: command is not supported 3: command timeout
		mac	The mac of BLE devices connected by gateway


```
{
      "msg_id":  3415,
      "device_info":  {
            "mac":  "4c11ae8be624"
      }
     "data":  {
            "mac":  "d9a5fa6f3882",
            "result_code":  0
      }
}
```


6.17 Report 3-axis data (416)

When the gateway enables the 3-axis data notify switch, the gateway will receive real time 3-axis data from the BXP-D device, then gateway will transfer to cloud.

Flag	Msg id offset	Parameter name	Parameter value
Notify	416	mac	string. 12 characters BLE device MAC address
		x_axis_data	number
		y_axis_data	number
		z_axis_data	number


```
{
      "msg_id":  3416,
      "device_info":  {
            "mac":  "4c11ae8be624"
      },
      "data":  {
            "mac":  "D9A5FA6F3882",
```

```
            "x_axis_data":  0,
            "y_axis_data":  50,
            "z_axis_data":  1500
      }
}
```

6.18 Set device power off (417)

To remote power off the beacon via gateway.

Flag	Msg id offset	Parameter name	Parameter value
Write	417	mac	The mac of BLE devices connected by gateway

setup:

```
{
      "msg_id":  1417,
      "device_info":  {
            "mac":  "4c11ae8be624"
      }
     "data":  {
            "mac":  "d9a5fa6f3882"
      }

}
```

Device reply:

```
{
      "msg_id":  1417,
      "device_info":  {
            "mac":  "4c11ae8be624"
      },
      "result_code":  0,
      "result_msg":  "success"
}
```

6.19 Report set device power off result (418)

When the gateway gets a reply from the BXP-D device for the power off command, it will report the result to cloud.

Flag	Msg id offset	Parameter name	Parameter value
Notify	418	mac	string. 12 characters BLE device MAC address

		result_code	number. 0-3 0: success 1: the BLE device is not connected by gateway 2: command is not supported 3: command timeout

```
{
      "msg_id":  3418,
      "device_info":  {
            "mac":  "4c11ae8be624"
      },
      "data":  {
            "mac":  "D9A5FA6F3882",
            "result_code":  0
      }
}
```

6.20 Get broadcast parameters (419)

To get the Bluetooth broadcast parameters of BLE device.

Flag	Msg id offset	Parameter name	Parameter value
Write	419	mac	string. 12 characters (6 bytes of BLE device mac address)

Setup:

```
{
      "msg_id":  1419,
      "device_info":  {
            "mac":  "4c11ae8be624"
      },
      "data":  {
            "mac":  "d9a5fa6f3882",
               }
}
```

6.21 Report broadcast parameters (420)

When the gateway gets the broadcast parameters from the BLE device, it will report to cloud.

Flag	Msg id offset	Parameter name	Parameter value
Notify	420	mac	string. 12 characters (6 bytes of BLE device mac address)
		result_code	number. 0-3 0: command success 1: the BLE device is not connected by gateway 2: command is not supported 3: command timeout
		adv_param	Array. The device has 6 adv slots, therefore it will have max 6 groups of advertising parameters. valid when result code is 0
		channel	number. 0~5
		channel_type	number. 0x00: Eddystone-UID 0x10: Eddystone-URL 0x20: Eddystone-TLM 0x30: Eddystone-EID 0x40: BXP-Device info 0x50: iBeacon 0x60: BXP-ACC 0x70: BXP-T&H 0xff: disable
		trigger_type	number. 0: no trigger 1: Temperature trigger 2: humidity trigger 3: double press trigger 4: Triple press trigger 5: Movement trigger 6: Light sensor trigger
		adv_interval	number. Unit: second
		tx_power	number. Unit: dBm


```
{
  "msg_id": 3420,
  "device_info": {
    "mac": "1c6920e45214"
  },
  "data": {
    "mac": "dd4b9c105f47",
```

```
    "result_code": 0,
    "adv_param": [
      "channel": 0,
      "channel_type": 1,
      "trigger_type": 0,
      "adv_interval": 1000,
      "tx_power": 0
        ]
  }
}
```

6.22 Set broadcast parameters (421)

To set Bluetooth broadcast parameters for BLE device.

Flag	Msg id offset	Parameter name	Parameter value
Write	421	mac	string. 12 characters (6 bytes of Button mac address)
		channel	number. 0~5
		adv_interval	number. 100~10000, unit: ms
		tx_power	number. (-40,-20,-16,-12,-8,-4,0,3,4), unit: dBm

Setup:

```
{
    "msg_id": 1421,
    "device_info": {
        "mac": "4c11ae8be624"
    },
    "data": {
        "mac": "d9a5fa6f3882",
        "channel": 1,
        "adv_interval": 100,
        "tx_power": 0
          }
}
```

6.23 Report broadcast parameters set result (422)

When the gateway gets a reply from the BXP-D Beacon for remote control power off command, it will notify the result to cloud.

Flag	Msg id offset	Parameter name	Parameter value
Notify	422	result_code	number. 0-3 0: command success 1: the Button is not connected by gateway 2: command is not supported 3: command timeout
		mac	string. 12 characters (6 bytes of BLE mac address)


```
{
      "msg_id": 3422,
      "device_info": {
            "mac": "4c11ae8be624"
      },
      "data": {
            "mac": "d9a5fa6f3882",
            "result_code": 0
               }
}
```


7. Connect and Manage BXP-T device

Send command from cloud to the gateway, to make the gateway connect and communicate with the BXP-Tag device.

7.1 Connect MOKO BXP-Tag device (450)

Send command from cloud to the gateway, to make the gateway connect with the BXP-Tag device.

Flag	Msg id offset	Parameter name	Parameter value
Write	450	mac	string. 12 characters (6 bytes of BLE device mac address)
		passwd	string. 0-16 characters (0-8 bytes of password).

Setup:


```
{
      "msg_id": 1450,
      "device_info": {
            "mac": "4c11ae8be624"
      },
```

```
    "data": {
        "mac": "D9A5FA6F3882",
        "passwd": "Moko4321"
    }
}
```

Reply:

```
{
    "msg_id": 1450,
    "device_info": {
        "mac": "4c11ae8be624"
    },
    "result_code": 0,
    "result_msg": "success"
}
```

7.2 Report connection result (451)

After the gateway finishes the connection, it will actively report the connection result to cloud. If connecting with BXP-T device successfully, the gateway will also inquire the device information from the device and transfer to cloud.

Flag	Msg id offset	Parameter name	Parameter value
Notify	451	mac	string. 12 characters (6 bytes of BLE device mac address)
		result_code	number. 0-8 0: Connect succeed 1: Gateway BLE scan disabled 2: Exceed the connectable number (the gateway is connected with an another BLE device) 3: BLE device connected (the BLE device has been connected with gateway) 4: BLE device out of range 5: BLE device un-connectable 6: Connect fail 7: BLE device type error(connection type error) 8: Password error
		result_msg	string. The description for the result code

		product_model	string. Valid when result code is 0
		company_name	string. Valid when result code is 0
		hardware_version	string. Valid when result code is 0
		software_version	string. Valid when result code is 0
		firmware_version	string. Valid when result code is 0
		sensor_status	Valid when result code is 0 number. Valid when result code is 0. ➢ Bit0: 3-axis sensor equipped status. 0- not equipped, 1- equipped. ➢ Bit1: temperature&humidity sensor equipped status. 0- not equipped, 1- equipped. Bit2: light sensor equipped status. 0- not equipped, 1- equipped.
		battery_v	number.Battery voltage, unit:mV valid when result code is 0


```
{
    "msg_id" : 3451,
  "device_info" : {
    "mac" : "142b2fe9b19c"
  },
  "data" : {
    "mac" : "a49e6965233b",
    "result_code" : 0,
    "result_msg" : "Connect succeed",
    "product_model" : "MK Tag",
    "company_name" : "MOKO TECHNOLOGY LTD.",
    "hardware_version" : "MKBSL series",
    "software_version" : "BXP-T-SLA",
    "firmware_version" : "V1.2.5",
    "sensor_status" : 0,
    "battery_v" : 3388
  }
}
```

7.3 Get BLE device information (452)

Send command from cloud to the gateway, to make the gateway inquire the device information of the BXP-T device.

Flag	Msg id offset	Parameter name	Parameter value
Write	452	mac	string. 12 characters (6 bytes of BLE device mac address)

Setup:


```
{
    "msg_id": 1452,
    "device_info": {
        "mac": "4c11ae8be624"
    },
    "data": {
        "mac": "D9A5FA6F3882"
    }
}
```


Reply:


```
{
    "msg_id": 1452,
    "device_info": {
        "mac": "4c11ae8be624"
    },
    "result_code": 0,
    "result_msg": "success"
}
```


7.4 Report BLE device information (453)

After the gateway gets device information from the BXP-T device, it will transfer to cloud.

Flag	Msg id offset	Parameter name	Parameter value
Notify	453	mac	string. 12 characters (6 bytes of BLE mac address)
		result_code	number. 0-2 0: success 1: the BLE device is not connected by gateway

}

			2: command is not supported
		product_model	string. Valid when the result code is 0
		company_name	string. Valid when the result code is 0
		hardware_version	string. Valid when the result code is 0
		software_version	string. Valid when the result code is 0
		firmware_version	string. Valid when the result code is 0
		sensor_status	Valid when result code is 0 number. Valid when result code is 0. ➢ Bit0: 3-axis sensor equipped status. 0- not equipped, 1- equipped. ➢ Bit1: temperature&humidity sensor equipped status. 0- not equipped, 1- equipped. Bit2: light sensor equipped status. 0- not equipped, 1- equipped.


```
{
    "msg_id": 3453,
    "device_info": {
        "mac": "4c11ae8be624"
    },
    "data": {
        "mac": "d9a5fa6f3882",
        "result_code": 0,
        "result_msg": "Connect succeed",
        "product_model": "BXP-T",
        "company_name": "MOKO TECHNOLOGY LTD.",
        "hardware_version": "MKBNseries",
        "software_version": "BXP-T",
        "firmware_version": "V1.0.1",
        "sensor_status": 3
    }
}
```

7.5 Get battery level (454)

To inquire the battery status of the BXP-Tag device via gateway.

Flag	Msg id offset	Parameter name	Parameter value
Write	454	mac	string. 12 characters (6 bytes of BLE device mac address)

Setup:


```
{
	"msg_id":  1454,
	"device_info":  {
		"mac":  "4c11ae8be624"
	},
	"data":  {
		"mac":  "D9A5FA6F3882"
	}
}
```


Reply:


```
{
	"msg_id":  1454,
	"device_info":  {
		"mac":  "4c11ae8be624"
	},
	"result_code":  0,
	"result_msg":  "success"
}
```


7.6 Report battery level (455)

After the gateway get battery level from the BXP-T device, it will report to cloud.

Flag	Msg id offset	Parameter name	Parameter value
Notify	455	mac	string. 12 characters (6 bytes of Button mac address)
		result_code	number. 0-2 0: success 1: the BLE device is not connected by gateway 2: command is not supported 3: command timeout
		battery_v	number, battery voltage , unit:

			mV Valid when the result code is 0


```
{
    "msg_id": 3455,
    "device_info": {
        "mac": "4c11ae8be624"
    },
    "data": {
        "mac": "d9a5fa6f3882",
        "result_code": 0,
        "battery_v": 3000
    }
}
```


7.7 Get 3-axis parameters (456)

To get the 3-axis parameters of BXP-T device via gateway.

Flag	Msg id offset	Parameter name	Parameter value
Write	456	mac	The mac of BLE devices connected by gateway

Setup:


```
{
    "msg_id": 1456,
    "device_info": {
        "mac": "4c11ae8be624"
    }
   "data": {
        "mac": "d9a5fa6f3882"
    }

}
```


Reply:


```
{
    "msg_id": 1456,
    "device_info": {
        "mac": "4c11ae8be624"
    },
    "result_code": 0,
    "result_msg": "success"
}
```

7.8 Report 3-axis parameters (457)

When the gateway gets a reply from the BXP-T device for the get 3-axis parameters command, it will report the result to cloud.

Flag	Msg id offset	Parameter name	Parameter value
Notify	457	mac	string. 12 characters BLE device MAC address
		result_code	number. 0-3 0: success 1: the BLE device is not connected by gateway 2: command is not supported 3: command timeout
		full_scale	number 0: 2g, unit of sensitivity is 3.91mg 1: 4g, unit of sensitivity is 7.81mg 2: 8g, unit of sensitivity is 15.63mg 3: 16g, unit of sensitivity is 31.25mg valid when result_code is 0
		sensitivity	number, 1-255 valid when result_code is 0
		sampling_rate	number 0:1hz 1:10hz 2:25hz 3:50hz 4:100hz valid when result_code is 0


```
{
	"msg_id":  3457,
	"device_info":  {
		"mac":  "4c11ae8be624"
	},
	"data":  {
		"mac":  "D9A5FA6F3882",
		"result_code":  0,
		"full_scale":  0,
		"sensitivity":  10,
		"sampling_rate":  1
```

```
    }
}
```

7.9 Set 3-axis parameters (458)

To set 3-axis parameters for BLE device.

Flag	Msg id offset	Parameter name	Parameter value
Write	458	mac	string. 12 characters BLE device MAC address
		full_scale	number. 0-3 0: success 1: the BLE device is not connected by gateway 2: command is not supported 3: command timeout
		sensitivity	number 0: 2g, unit of sensitivity is 3.91mg 1: 4g, unit of sensitivity is 7.81mg 2: 8g, unit of sensitivity is 15.63mg 3: 16g, unit of sensitivity is 31.25mg valid when result_code is 0
		sampling_rate	number, 1-255 valid when result_code is 0
			number 0:1hz 1:10hz 2:25hz 3:50hz 4:100hz valid when result_code is 0

Setup:

```
{
  "msg_id": 1458,
  "device_info": {
    "mac": "1c6920e45214"
  },
  "data": {
    "mac": "6cfd2276246a",
    "full_scale": 0,
    "sensitivity": 20,
```

```
    "sampling_rate": 1
  }
}
```

7.10 Report 3-axis parameters set result (459)

After the gateway get reply from the BLE device for the 3-axis parameters set command, it will report to cloud.

Flag	Msg id offset	Parameter name	Parameter value
Notify	459	mac	string. 12 characters (6 bytes of BLE device mac address)
		result_code	number. 0-3 0: success 1: the Button is not connected by gateway 2: command is not supported 3: command timeout

```
{
    "msg_id": 3459,
    "device_info": {
        "mac": "4c11ae8be624"
    },
    "data": {
        "mac": "d9a5fa6f3882",
        "result_code": 0
    }
}
```

7.11 Get movement trigger count (460)

To get movement trigger count of BLE device via gateway.

Flag	Msg id offset	Parameter name	Parameter value
Write	460	mac	string. 12 characters (6 bytes of BLE device mac address)

Setup:

```
{
    "msg_id": 1460,
    "device_info": {
```

```
        "mac": "4c11ae8be624"
    },
    "data": {
        "mac": "d9a5fa6f3882"
          }
}
```


7.12 Report movement trigger count (461)

When the gateway gets a reply from the BXP-Tag device for get movement trigger count command, it will report to cloud.

Flag	Msg id offset	Parameter name	Parameter value
Notify	461	mac	string. 12 characters (6 bytes of BLE mac address)
		result_code	number. 0-3 0: success 1: the BLE device is not connected by gateway 2: command is not supported 3: command timeout
		count	number.


```
{
    "msg_id": 3461,
    "device_info": {
        "mac": "4c11ae8be624"
    },
    "data": {
        "mac": "d9a5fa6f3882",
        "result_code": 0,
        "count": 100
          }
}
```


7.13 Clear movement trigger count (462)

To clear movement trigger count of the BLE device via gateway.

Flag	Msg id offset	Parameter name	Parameter value
Write	462	mac	string. 12 characters (6 bytes of BLE mac address)

Setup:


```
{
    "msg_id": 1462,
    "device_info": {
        "mac": "4c11ae8be624"
    },
    "data": {
        "mac": "d9a5fa6f3882"
        }
}
```


7.14 Report movement trigger count clear result (463)

When the gateway gets a reply from the BXP-T device for clear movement trigger count command, it will report the result to cloud.

Flag	Msg id offset	Parameter name	Parameter value
Notify	463	mac	string. 12 characters (6 bytes of BLE mac address)
		result_code	number. 0-3 0: command success 1: the BLE device is not connected by gateway 2: command is not supported 3: command timeout


```
{
    "msg_id": 3463,
    "device_info": {
        "mac": "4c11ae8be624"
    },
    "data": {
        "mac": "D9A5FA6F3882",
        "result_code": 0
    }
}
```


7.15 Setup LED (464)

To setup the LED of BXP-Tag device.

Flag	Msg id offset	Parameter name	Parameter value

Write	464	mac	string. 12 characters (6 bytes of BLE mac address)
		color	string. green blue red
		flash_interval	number. 100-10000, unit:ms
		flash_time	number. 1-600, unit: s

Setup:


```
{
	"msg_id": 1464,
	"device_info": {
		"mac": "4c11ae8be624"
	},
	"data": {
		"mac": "d9a5fa6f3882",
		"color": “green”,
		"flash_interval": 500,
		"flash_time": 2
	}
}
```


7.16 Report LED set result (465)

When the gateway gets a reply from the BXP-Tag device for setup LED command, it will report the result to cloud.

Flag	Msg id offset	Parameter name	Parameter value
Notify	465	result_code	number. 0-3 0: command success 1: the BLE device is not connected by gateway 2: command is not supported 3: command timeout
		mac	The mac of BLE devices connected by gateway

Setup:


```
{
 "msg_id": 3465,
	"device_info": {
```

```
        "mac": "4c11ae8be624"
    }
    "data": {
        "mac": "d9a5fa6f3882",
        "result_code": 0
    }

}
```

7.17 Set 3-axis data notify switch (466)

To setup the BXP-T device to enable/ disable 3-axis data notify switch.

Flag	Msg id offset	Parameter name	Parameter value
Write	466	switch_value	number. 0-1 0: disable 1: enable
		mac	The mac of BLE device connected by gateway

Setup:

```
{
    "msg_id": 1466,
    "device_info": {
        "mac": "4c11ae8be624"
    },
    "data": {
        "mac": "d9a5fa6f3882",
        "switch_value": 1
    }
}
```

7.18 Report 3-axis data notify switch set result(467)

When the gateway gets a reply from the MK Button device for the 3-axis notify switch setup command, it will report the result to cloud.

Flag	Msg id offset	Parameter name	Parameter value
Notify	467	mac	string. 12 characters BLE device MAC address
		result_code	number. 0-3 0: command success

			1: the BLE device is not connected with gateway 2: command is not supported 3: command timeout


```
{
    "msg_id": 3467,
    "device_info": {
        "mac": "4c11ae8be624"
    }
   "data": {
        "mac": "d9a5fa6f3882",
        "result_code": 0
    }

}
```


7.19 Report 3-axis data (468)

When the gateway enables the 3-axis data notify switch, it will receive real time 3-axis data from the BXP-T device, then the gateway will transfer 3-axis data to cloud.

Flag	Msg id offset	Parameter name	Parameter value
Notify	468	mac	string. 12 characters BLE device MAC address
		x_axis_data	number, unit: mg
		y_axis_data	number, unit: mg
		z_axis_data	number, unit: mg


```
{
    "msg_id": 3468,
    "device_info": {
        "mac": "4c11ae8be624"
    },
    "data": {
        "mac": "D9A5FA6F3882",
        "x_axis_data": 0,
        "y_axis_data": 50,
        "z_axis_data": 1500
    }
}
```

7.20 Set device power off (469)

To power off the beacon via gateway.

Flag	Msg id offset	Parameter name	Parameter value
Write	469	mac	The mac of BLE devices connected by gateway

setup:


```
{
      "msg_id":  1469,
      "device_info":  {
            "mac":  "4c11ae8be624"
      }
     "data":  {
            "mac":  "d9a5fa6f3882"
      }

}
```


7.21 Report set device power off result(470)

When the gateway gets a reply from the BXP-T device for the power off set command, it will report the result to cloud.

Flag	Msg id offset	Parameter name	Parameter value
Notify	470	mac	string. 12 characters BLE device MAC address
		result_code	number. 0-3 0: command success 1: the BLE device is not connected by gateway 2: command is not supported 3: command timeout


```
{
      "msg_id":  3470,
      "device_info":  {
            "mac":  "4c11ae8be624"
      },
      "data":  {
            "mac":  "D9A5FA6F3882",
            "result_code":  0
```

```
    }
}
```

7.22 Get broadcast parameters (471)

To inquire Bluetooth broadcast parameters of BXP-T device.

Flag	Msg id offset	Parameter name	Parameter value
Write	471	mac	string. 12 characters (6 bytes of BLE device mac address)

Setup:

```
{
    "msg_id": 1471,
    "device_info": {
        "mac": "4c11ae8be624"
    },
    "data": {
        "mac": "d9a5fa6f3882",
          }
}
```

7.23 Report broadcast parameters (472)

When the gateway gets a reply from the BLE device for the get Bluetooth broadcast parameters command, it will report to cloud.

Flag	Msg id offset	Parameter name	Parameter value
Notify	472	mac	string. 12 characters (6 bytes of BLE device mac address)
		result_code	number. 0-3 0: command success 1: the BLE device is not connected by gateway 2: command is not supported 3: command timeout
		adv_param	Array. valid when result code is 0. The BLE device has six adv channels, therefore it has at most 6 groups of adv parameters.
		channel	number. 0-5
		channel_type	number.

			0(0x00): UID 16(0X10): URL 32(0X20): TLM 90(0X50): iBeacon 128(0X80): Tag info 255(0XFF): no data
		trigger_type	number. 0: no trigger 5: movement trigger 6: hall trigger
		adv_interval	number. 1-100, unit:100ms
		tx_power	number.


```
{
    "msg_id" : 3472,
    "device_info" : {
    "mac" : "142b2fe9b19c"
  },
  "data" : {
    "mac" : "a49e6965233b",
    "result_code" : 0,
    "adv_param" : [ {
      "channel" : 0,
      "channel_type" : 128,
      "trigger_type" : 0,
      "adv_interval" : 10,
      "tx_power" : 0
    }, {
      "channel" : 1,
      "channel_type" : 32,
      "trigger_type" : 0,
      "adv_interval" : 10,
      "tx_power" : 0
    }, {
      "channel" : 2,
      "channel_type" : 255
    }, {
      "channel" : 3,
      "channel_type" : 255
    }, {
      "channel" : 4,
      "channel_type" : 255
    }, {
      "channel" : 5,
```

```
        "channel_type" : 255
      } ]
   }
}
```

7.24 Set broadcast parameters (473)

To set beacon Bluetooth broadcast parameters for BXP-T device via gateway.

Flag	Msg id offset	Parameter name	Parameter value
Write	473	mac	string. 12 characters (6 bytes of Button mac address)
		channel	number. 0-5
		adv_interval	number. 1-100 unit:100ms
		tx_power	number. {-20,-16,-12,-8,-4,0,3,4,6}

Setup:

```
{
     "msg_id": 1473,
     "device_info": {
          "mac": "4c11ae8be624"
     },
     "data": {
          "mac": "d9a5fa6f3882",
          "channel": 1,
          "adv_interval": 100,
          "tx_power": 0
            }
}
```

7.25 Report broadcast parameters set result (474)

When the gateway gets reply from the BLE device for the Bluetooth broadcast parameters set command, it will report the result to cloud.

Flag	Msg id offset	Parameter name	Parameter value
Notify	474	result_code	number. 0-3 0: command success 1: the BLE device is not connected by gateway 2: command is not supported 3: command timeout
		mac	string. 12 characters (6 bytes of

			BLE device mac address)


```
{
    "msg_id": 3474,
    "device_info": {
        "mac": "4c11ae8be624"
    },
    "data": {
        "mac": "d9a5fa6f3882",
        "result_code": 0
        }
}
```


8. Connect and Manage BXP-S device

Send command from cloud to the gateway, to make the gateway connect and communicate with the BXP-Sensor device.

8.1 Connect MOKO BXP-S device (500)

Send command from cloud to the gateway, to make the gateway connect and communicate with the BXP-Sensor device.

Flag	Msg id offset	Parameter name	Parameter value
Write	500	mac	string. 12 characters (6 bytes of BLE device mac address)
		passwd	string. 0-16 characters (0-8 bytes of password).

Setup:


```
{
    "msg_id": 1500,
    "device_info": {
        "mac": "4c11ae8be624"
    },
    "data": {
        "mac": "D9A5FA6F3882",
        "passwd": "Moko4321"
```

```
    }
}
```
Reply:

```
{
    "msg_id": 1500,
    "device_info": {
        "mac": "4c11ae8be624"
    },
    "result_code": 0,
    "result_msg": "success"
}
```


8.2 Report connection result (501)

After the gateway finishes the connection, it will actively report the connection result to cloud. If connecting with BXP-sensor device successfully, the gateway will also inquire the device information from the device and transfer to cloud.

Flag	Msg id offset	Parameter name	Parameter value
Notify	501	mac	string. 12 characters (6 bytes of BLE device mac address)
		result_code	number. 0-8 0: Connect succeed 1: Gateway BLE scan disabled 2: Exceed the connectable number (the gateway is connected with an another BLE device) 3: BLE device connected (the BLE device has been connected with gateway) 4: BLE device out of range 5: BLE device un-connectable 6: Connect fail 7: BLE device type error(connection type error) 8: Password error
		result_msg	string. The description for the result code
		product_model	string. Valid when result code is 0
		company_name	string.

			Valid when result code is 0
		hardware_version	string. Valid when result code is 0
		software_version	string. Valid when result code is 0
		firmware_version	string. Valid when result code is 0
		axis_type	Valid when result code is 0 0: no 3-axis sensor 1-2: has 3-axis sensor
		th_type	Valid when result code is 0 0: no T&H sensor 1-4: has T&H sensor
		light_type	Valid when result code is 0 0: no light sensor 1: has light sensor
		pir_type	Valid when result code is 0 0: no pir sensor 1: has pir sensor
		tof_type	Valid when result code is 0 0: no tof sensor 1: has tof sensor
		battery_level	number. unit: % Valid when result code is 0
		battery_v	**number.** **Battery voltage, unit:mV** **Valid when result code is 0** **Added in gateway V2.X.X firmware.**


```
{
  "msg_id" : 3501,
  "device_info" : {
    "mac" : "142b2fe9b19c"
  },
  "data" : {
    "mac" : "5cc7c1f836f6",
    "result_code" : 0,
    "result_msg" : "Connect succeed",
    "product_model" : "MK Sensor",
    "company_name" : "MOKO TECHNOLOGY LTD.",
    "hardware_version" : "MKBSL series",
    "software_version" : "BXP-S-SLATHF",
    "firmware_version" : "V1.0.4",
```

```
        "axis_type" : 1,
        "th_type" : 3,
        "light_type" : 0,
        "pir_type" : 0,
        "tof_type" : 0,
        "battery_level" : 62,
        "battery_v" : 2890
    }
}
```

8.3 Get BLE device information (502)

Send command from cloud to the gateway, to make the gateway inquire the device information of the BXP-S device.

Flag	Msg id offset	Parameter name	Parameter value
Write	502	mac	string. 12 characters (6 bytes of BLE device mac address)

Setup:

```
{
        "msg_id": 1502,
        "device_info": {
                "mac": "4c11ae8be624"
        },
        "data": {
                "mac": "D9A5FA6F3882"
        }
}
```

Reply:

```
{
        "msg_id": 1502,
        "device_info": {
                "mac": "4c11ae8be624"
        },
        "result_code": 0,
        "result_msg": "success"
}
```

8.4 Report BLE device information (503)

After the gateway gets device information from the BXP-S device, it will transfer to cloud.

Flag	Msg id offset	Parameter name	Parameter value
Notify	503	mac	string. 12 characters (6 bytes of BLE mac address)
		result_code	number. 0-2 0: success 1: the BLE device is not connected by gateway 2: command is not supported
		product_model	string. Valid when result code is 0
		company_name	string. Valid when result code is 0
		hardware_version	string. Valid when result code is 0
		software_version	string. Valid when result code is 0
		firmware_version	string. Valid when result code is 0
		axis_type	Valid when result code is 0 0: no 3-axis sensor 1-2: has 3-axis sensor
		th_type	Valid when result code is 0 0: no T&H sensor 1-4: has T&H sensor
		light_type	Valid when result code is 0 0: no light sensor 1: has light sensor
		pir_type	Valid when result code is 0 0: no pir sensor 1: has pir sensor
		tof_type	Valid when result code is 0 0: no tof sensor 1: has tof sensor


```
{
    "msg_id": 3503,
    "device_info": {
        "mac": "4c11ae8be624"
```

```
    },
    "data": {
    "mac" : "5cc7c1f836f6",
    "result_code" : 0,
    "result_msg" : "Connect succeed",
    "product_model" : "MK Sensor",
    "company_name" : "MOKO TECHNOLOGY LTD.",
    "hardware_version" : "MKBSL series",
    "software_version" : "BXP-S-SLATHF",
    "firmware_version" : "V1.0.4",
    "axis_type" : 1,
    "th_type" : 3,
    "light_type" : 0,
    "pir_type" : 0,
    "tof_type" : 0
    }
}
```

8.5 Get battery info (504)

To inquire the battery status of the BLE device via gateway.

Flag	Msg id offset	Parameter name	Parameter value
Write	504	mac	string. 12 characters (6 bytes of BLE device mac address)

Setup:

```
{
    "msg_id": 1504,
    "device_info": {
        "mac": "4c11ae8be624"
    },
    "data": {
        "mac": "D9A5FA6F3882"
    }
}
```

Reply:

```
{
    "msg_id": 1504,
    "device_info": {
        "mac": "4c11ae8be624"
    },
    "result_code": 0,
```

```
    "result_msg": "success"
}
```


8.6 Report battery info (505)

After the gateway get battery level from the BLE device, it will report to cloud.

Flag	Msg id offset	Parameter name	Parameter value
Notify	505	mac	string. 12 characters (6 bytes of Button mac address)
		result_code	number. 0-2 0: success 1: the BLE device is not connected by gateway 2: command is not supported 3: command timeout
		battery_level	number, battery percentage , unit: % Valid when the result code is 0
		battery_v	**number.** **Battery voltage, unit:mV** **Valid when result code is 0** **Added in gateway V2.X.X firmware.**


```
{
    "msg_id": 3505,
    "device_info": {
        "mac": "4c11ae8be624"
    },
    "data": {
        "mac": "d9a5fa6f3882",
        "result_code": 0,
        "battery_level": 89,
         "battery_v" : 2890
    }
}
```


8.7 Set real-time T&H data notify switch (506)

To enable/ disable the real-time temperature &humidity notify switch.

Flag	Msg id offset	Parameter name	Parameter value
Write	506	switch_value	number. 0: disable 1: enable
		mac	string. 12 characters (6 bytes of BLE device mac address)

Setup:


```
{
      "msg_id": 1506,
      "device_info": {
            "mac": "4c11ae8be624"
      }
     "data": {
            "mac": "d9a5fa6f3882",
            "switch_value": 1
      }
}
```


Reply:


```
{
      "msg_id": 1506,
      "device_info": {
            "mac": "4c11ae8be624"
      },
      "result_code": 0,
      "result_msg": "success"
}
```


8.8 Report real-time T&H data notify switch set result (507)

After the gateway gets reply for the T&H data notify switch set command, it will report the result to cloud.

Flag	Msg id offset	Parameter name	Parameter value
Notify	507	mac	string. 12 characters (6 bytes of BLE device mac address)
		result_code	number. 0-2 0: success 1: the BLE device is not connected by gateway 2: command is not supported

			3: command time out


```
{
    "msg_id":  3507,
    "device_info":  {
        "mac":  "4c11ae8be624"
    },
    "data":  {
        "mac":  "d9a5fa6f3882",
        "result_code":  0
    }
}
```


8.9 Report real-time T&H data (508)

When the gateway enables the real time temperature and humidity data notify switch, the gateway will receive real time temperature and humidity data from the device, then gateway will transfer to cloud.

Flag	Msg id offset	Parameter name	Parameter value
Notify	508	mac	string. 12 characters (6 bytes of BLE device mac address)
		temperature	number, unit ℃
		humidity	number. unit %


```
{
  "msg_id"  :  3508,
  "device_info"  :  {
    "mac"  :  "142b2fe9b19c"
  },
  "data"  :  {
    "mac"  :  "e406bfa7322a",
    "temperature"  :  19.5,
    "humidity"  :  46.3
  }
}
```


8.10 Set 3-axis data notify switch (509)

To setup the BXP-S device to enable/ disable 3-axis data notify switch.

Flag	Msg id offset	Parameter name	Parameter value

Write	509	switch_value	number. 0-1 0: disable 1: enable
		mac	The mac of BLE device connected by gateway

Setup:


```
{
    "msg_id": 1509,
    "device_info": {
        "mac": "4c11ae8be624"
    },
    "data": {
        "mac": "d9a5fa6f3882",
        "switch_value": 1
    }
}
```


8.11 Report 3-axis data notify switch set result (510)

When the gateway gets a reply from the BLE device for the 3-axis notify switch setup command, it will report the result to cloud.

Flag	Msg id offset	Parameter name	Parameter value
Notify	510	mac	string. 12 characters BLE device MAC address
		result_code	number. 0-3 0: command success 1: the BLE device is not connected with gateway 2: command is not supported 3: command timeout


```
{
    "msg_id": 3510,
    "device_info": {
        "mac": "4c11ae8be624"
    }
   "data": {
        "mac": "d9a5fa6f3882",
        "result_code": 0
    }
}
```

8.12 Report 3-axis data (511)

When the gateway enables the 3-axis data notify switch, the gateway will receive real time 3-axis data from the BXP-S device, then gateway will transfer to cloud.

Flag	Msg id offset	Parameter name	Parameter value
Notify	511	mac	string. 12 characters BLE device MAC address
		x_axis_data	number,mg
		y_axis_data	number,mg
		z_axis_data	number,mg


```
{
      "msg_id":  3511,
      "device_info":  {
            "mac":  "4c11ae8be624"
      },
      "data":  {
            "mac":  "D9A5FA6F3882",
            "x_axis_data":  0,
            "y_axis_data":  50,
            "z_axis_data":  1500
      }
}
```


8.13 Set history T&H data notify switch (512)

To set the history temperature &humidity notify switch for BXP-S device via gateway.

Flag	Msg id offset	Parameter name	Parameter value
Write	512	switch_value	number. 0: disable 1: enable
		mac	string. 12 characters (6 bytes of BLE device mac address)

Setup:


```
{
      "msg_id":  1512,
      "device_info":  {
```

"mac": "4c11ae8be624"
}
"data": {
"mac": "d9a5fa6f3882"
}
}
Reply:
{
"msg_id": 1512,
"device_info": {
"mac": "4c11ae8be624"
},
"result_code": 0,
"result_msg": "success"
}

8.14 Report history T&H data notify switch set result (513)

After the gateway gets reply for the history T&H data notify switch set command, it will report the result to cloud.

Flag	Msg id offset	Parameter name	Parameter value
Notify	513	result_code	number. 0-3 0: success 1: the BLE device is not connected by gateway 2: command is not supported 3: command timeout
		mac	string. 12 characters (6 bytes of BLE mac address)

{
"msg_id": 3513,
"device_info": {
"mac": "4c11ae8be624"
},
"data": {
"mac": "d9a5fa6f3882",
"result_code": 0
}
}

8.15 Report history T&H data (514)

When the gateway enables the history temperature and humidity notify switch, the gateway will receive history temperature and humidity data from the device, then gateway will transfer to cloud.

Flag	Msg id offset	Parameter name	Parameter value
Notify	514	mac	string. 12 characters (6 bytes of Button mac address)
		history	object
		timestamp	number. unit: second
		temperature	number, unit ℃
		humidity	number. unit %


```
{
   "msg_id" : 3514,
   "device_info" : {
      "mac" : "142b2fe9b19c"
   },
   "data" : {
      "mac" : "e406bfa7322a",
      "history" : [ {
         "timestamp" : 1770948303,
         "temperature" : 25.6,
         "humidity" : 38.2
      }, {
         "timestamp" : 1770948603,
         "temperature" : 29.3,
         "humidity" : 49.2
      }, {
         "timestamp" : 1770948903,
         "temperature" : 23.2,
         "humidity" : 40.4
      }, {
         "timestamp" : 1770949203,
         "temperature" : 21.4,
         "humidity" : 43.4
      }, {
         "timestamp" : 1770949503,
         "temperature" : 19.7,
         "humidity" : 46.7
      }, {
         "timestamp" : 1770949803,
```

```
        "temperature" : 19.3,
        "humidity" : 46.4
    }, {
        "timestamp" : 1770950103,
        "temperature" : 19.6,
        "humidity" : 47.3
    }, {
        "timestamp" : 1770950403,
        "temperature" : 19.7,
        "humidity" : 45.8
    }, {
        "timestamp" : 1770950703,
        "temperature" : 19.8,
        "humidity" : 45.8
    }, {
        "timestamp" : 1770951003,
        "temperature" : 19.8,
        "humidity" : 45.8
    }, {
        "timestamp" : 1770951303,
        "temperature" : 19.7,
        "humidity" : 46.0
    } ]
  }
}
```

8.16 Clear history T&H data (515)

To clear history temperature &humidity data from BXP-S device.

Flag	Msg id offset	Parameter name	Parameter value
Write	515	mac	string. 12 characters (6 bytes of BLE device mac address)

Setup:

```
{
    "msg_id": 1515,
    "device_info": {
        "mac": "4c11ae8be624"
    },
    "data": {
        "mac": "d9a5fa6f3882"
    }
```

```
}
```

8.17 Notify clear history T&H data result (516)

When the gateway gets a reply from the BXP-S device for the delete history temperature and humidity command, it will transfer to cloud.

Flag	Msg id offset	Parameter name	Parameter value
Notify	516	mac	string. 12 characters (6 bytes of BLE device mac address)
		result_code	number. 0-3 0: success 1: the BLE device is not connected by gateway 2: command is not supported 3: command timeout


```
{
	"msg_id": 3516,
	"device_info": {
		"mac": "4c11ae8be624"
	},
	"data": {
		"mac": "d9a5fa6f3882",
		"result_code": 0
		  }
}
```


8.18 Get T&H sensor sampling rate(517)

To get temperature &humidity sensor parameters of BXP-S device via gateway.

Flag	Msg id offset	Parameter name	Parameter value
Write	517	mac	string. 12 characters (6 bytes of BLE device mac address)

Setup:


```
{
	"msg_id": 1517,
	"device_info": {
		"mac": "4c11ae8be624"
	},
	"data": {
```

```
            "mac":  "d9a5fa6f3882"
        }
}
```

8.19 Report T&H sensor sampling rate(518)

When the gateway gets a reply from the BXP-S device for the getting temperature &humidity sampling rate command, it will notify the result to cloud.

Flag	Msg id offset	Parameter name	Parameter value
Write	518	result_code	number. 0-3 0: command success 1: the ble device is not connected by gateway 2: command is not supported 3: command timeout
		sampling_rate	number, 1-65535, unit: second valid when result_code is 0
		mac	string. 12 characters BLE device MAC address

```
{
      "msg_id":  3518,
      "device_info":  {
            "mac":  "4c11ae8be624"
      },
      "data":  {
            "mac":  "D9A5FA6F3882",
            "sampling_rate":  10
      }
}
```

8.20 Set T&H sensor sampling rate (519)

To set temperature &humidity sensor sampling rate for BXP-S device.

Flag	Msg id offset	Parameter name	Parameter value
Write	519	mac	string. 12 characters (6 bytes of BLE device mac address)
		sampling_rate	number. 1-65535, unit: second

Setup:

```
{
```

```
    "msg_id": 1519,
    "device_info": {
        "mac": "4c11ae8be624"
    },
    "data": {
        "mac": "d9a5fa6f3882",
        "sampling_rate": 10
    }
}
```

8.21 Report T&H sensor sampling rate set result (520)

When the gateway gets a reply from the BXP-S Beacon for set temperature &humidity sensor parameters command, it will notify the result to cloud.

Flag	Msg id offset	Parameter name	Parameter value
Notify	520	result_code	number. 0-3 0: command success 1: the BLE device is not connected by gateway 2: command is not supported 3: command timeout
		mac	string. 12 characters BLE device MAC address

```
{
    "msg_id": 3520,
    "device_info": {
        "mac": "4c11ae8be624"
    }
   "data": {
        "mac": "d9a5fa6f3882",
        "result_code": 0
    }
}
```

8.22 Get hall sensor trigger count (521)

To get hall sensor trigger count of the BXP-S device via gateway.

Flag	Msg id offset	Parameter name	Parameter value
Write	521	mac	string. 12 characters (6 bytes of BLE device mac address)

Setup:


```
{
    "msg_id": 1521,
    "device_info": {
        "mac": "4c11ae8be624"
    },
    "data": {
        "mac": "d9a5fa6f3882"
    }
}
```


8.23 Report hall sensor trigger count (522)

When the gateway gets a reply from the BXP-S device for the get hall sensor trigger count command, it will report to cloud.

Flag	Msg id offset	Parameter name	Parameter value
Notify	522	mac	The mac of BLE devices connected by gateway
		result_code	number. 0-3 0: command success 1: the BLE device is not connected by gateway 2: command is not supported 3: command timeout
		num	number.


```
{
    "msg_id": 3522,
    "device_info": {
        "mac": "4c11ae8be624"
    }
    "data": {
        "mac": "d9a5fa6f3882",
        "num": 20
    }
}
```

8.24 Clear hall sensor trigger count (523)

To clear hall sensor trigger count from the BXP-S device.

Flag	Msg id offset	Parameter name	Parameter value
Write	523	mac	string. 12 characters (6 bytes of BLE device mac address)

Setup:


```
{
	"msg_id": 1523,
	"device_info": {
		"mac": "4c11ae8be624"
	},
	"data": {
		"mac": "d9a5fa6f3882"
		}
}
```


8.25 Report hall trigger count clear result (524)

When the gateway gets a reply from the BXP-S device for clear movement trigger count command, it will report the result to cloud.

Flag	Msg id offset	Parameter name	Parameter value
Write	524	mac	string. 12 characters (6 bytes of BLE mac address)
		result_code	number. 0-3 0: command success 1: the BLE device is not connected by gateway 2: command is not supported 3: command timeout

Setup:


```
{
	"msg_id": 3524,
	"device_info": {
		"mac": "4c11ae8be624"
	},
	"data": {
		"mac": "D9A5FA6F3882",
```

```
        "result_code": 0
    }
}
```


8.26 Setup LED (525)

To setup the LED of BXP-S device.

Flag	Msg id offset	Parameter name	Parameter value
Write	525	mac	string. 12 characters (6 bytes of BLE mac address)
		color	string. green blue red
		flash_interval	number. 100-10000, unit:ms
		flash_time	number. 10-6000, unit: 100ms

Setup:


```
{
    "msg_id": 1525,
    "device_info": {
        "mac": "4c11ae8be624"
    },
    "data": {
        "mac": "d9a5fa6f3882",
        "color": "green",
        "flash_interval": 500,
        "flash_time": 20
    }
}
```


8.27 Report LED set result (526)

When the gateway gets a reply from the BXP-S device for setup LED command, it will report the result to cloud.

Flag	Msg id offset	Parameter name	Parameter value
Notify	526	result_code	number. 0-3 0: command success 1: the BLE device is not connected by gateway

			2: command is not supported 3: command timeout
		mac	The mac of BLE devices connected by gateway


```
{
      "msg_id": 3526,
      "device_info": {
            "mac": "4c11ae8be624"
      }
     "data": {
            "mac": "d9a5fa6f3882",
            "result_code": 0
      }

}
```


8.28 Set device power off (527)

To power off the beacon via gateway.

Flag	Msg id offset	Parameter name	Parameter value
Write	527	mac	The mac of BLE devices connected by gateway

Setup:


```
{
      "msg_id": 1527,
      "device_info": {
            "mac": "4c11ae8be624"
      }
     "data": {
            "mac": "d9a5fa6f3882"
      }

}
```


8.29 Report set device power off result(528)

When the gateway gets a reply from the BXP-S device for the power off set command, it will report the result to cloud.

Flag	Msg id offset	Parameter name	Parameter value
Notify	528	mac	string. 12 characters BLE device MAC address
		result_code	number. 0-3 0: command success 1: the BLE device is not connected by gateway 2: command is not supported 3: command timeout


```
{
    "msg_id": 3528,
    "device_info": {
        "mac": "4c11ae8be624"
    },
    "data": {
        "mac": "D9A5FA6F3882",
        "result_code": 0
    }
}
```


8.31 Get broadcast parameters (529)

To inquire Bluetooth broadcast parameters of BXP-S device.

Flag	Msg id offset	Parameter name	Parameter value
Write	529	mac	string. 12 characters (6 bytes of BLE device mac address)

Setup:


```
{
    "msg_id": 1529,
    "device_info": {
        "mac": "4c11ae8be624"
    },
    "data": {
        "mac": "d9a5fa6f3882",
         }
}
```

8.32 Report broadcast parameters (530)

When the gateway gets a reply from the BXP-S device for get Bluetooth broadcast parameters command, it will report to cloud.

Flag	Msg id offset	Parameter name	Parameter value
Notify	529	mac	string. 12 characters (6 bytes of BLE device mac address)
		result_code	number. 0-3 0: command success 1: the BLE device is not connected by gateway 2: command is not supported 3: command timeout
		adv_param	Array. valid when result code is 0. The BLE device has 3 adv channels, therefore it has at most 3 groups of adv parameters.
		channel	number. 0-2
		enable	number. 0/1
		channel_type	number. 0: normal adv 1: trigger-after adv 6: trigger-before and trigger-after adv
		normal_adv	Object. Normal adv parameters
		adv_type	number. 0(0x00): UID 16(0X10): URL 32(0X20): TLM 90(0X50): iBeacon 112(0X70): TH_info 128(0X80): Tag info 144(0X90): no data
		adv_interval	number. unit:ms
		tx_power	number.
		trigger_before_adv	Object. Before trigger adv parameters
		adv_type	number. 0(0x00): UID

			16(0X10): URL 32(0X20): TLM 90(0X50): iBeacon 112(0X70): TH_info 128(0X80): Tag info 144(0X90): no data
		adv_interval	number, unit:ms
		tx_power	number.
		trigger_after_adv	Object. After trigger adv parameters
		adv_type	number. 0(0x00): UID 16(0X10): URL 32(0X20): TLM 90(0X50): iBeacon 112(0X70): TH_info 128(0X80): Tag info 144(0X90): no data
		adv_interval	number, unit:ms
		tx_power	number.


```
{
  "msg_id" : 3530,
  "device_info" : {
    "mac" : "142b2fe9b19c"
  },
  "data" : {
    "mac" : "e406bfa7322a",
    "result_code" : 0,
    "adv_param" : [ {
      "channel" : 0,
      "enable" : 1,
      "channel_type" : 0,
      "normal_adv" : {
        "adv_type" : 32,
        "adv_interval" : 1000,
        "tx_power" : 0
      }
    }, {
      "channel" : 1,
      "enable" : 1,
      "channel_type" : 2,
      "trigger_before_adv" : {
```

```
            "adv_type" : 128,
            "adv_interval" : 5000,
            "tx_power" : 0
        },
        "trigger_after_adv" : {
            "adv_type" : 128,
            "adv_interval" : 1000,
            "tx_power" : 4
        }
    }, {
        "channel" : 2,
        "enable" : 0
    } ]
  }
}
```

8.33 Set broadcast parameters (531)

To set Bluetooth broadcast parameters for BXP-S device via gateway.

Flag	Msg id offset	Parameter name	Parameter value
Write	531	mac	string. 12 characters (6 bytes of BLE mac address)
		channel	number. 0-2
		normal_adv	Object. Normal adv parameters
		adv_interval	number. 20-65535, unit:ms
		tx_power	number. {-20,-16,-12,-8,-4,0,3,4,6}
		trigger_before_adv	Object. Before trigger adv parameters
		adv_interval	number. 20-65535, unit:ms
		tx_power	number. {-20,-16,-12,-8,-4,0,3,4,6}
		trigger_after_adv	Object. After trigger adv parameters
		adv_interval	number. 20-65535, unit:ms
		tx_power	number. {-20,-16,-12,-8,-4,0,3,4,6}

Setup:

```
{
    "msg_id": 1531,
```

```
    "device_info": {
        "mac": "4c11ae8be624"
    },
      "data" : {
      "mac" : "e406bfa7322a",
      "channel" : 1,
      "trigger_before_adv" : {
      "adv_interval" : 8000,
      "tx_power" : 0
      },
      "trigger_after_adv" : {
      "adv_interval" : 1000,
      "tx_power" : 4
      }
  }
}
```

8.34 Report broadcast parameters set result (532)

When the gateway gets reply from the BLE device for the Bluetooth broadcast parameters set command, it will report the result to cloud.

Flag	Msg id offset	Parameter name	Parameter value
Notify	532	result_code	number. 0-3 0: command success 1: the BLE device is not connected by gateway 2: command is not supported 3: command timeout
		mac	string. 12 characters (6 bytes of BLE device mac address)

```
{
    "msg_id": 3532,
    "device_info": {
        "mac": "4c11ae8be624"
    },
    "data": {
        "mac": "d9a5fa6f3882",
        "result_code": 0
          }
}
```

8.35 Setup Buzzer (533)

To setup the buzzer of BXP-S device.

Flag	Msg id offset	Parameter name	Parameter value
Write	533	mac	string. 12 characters (6 bytes of BLE mac address)
		ring_time	Buzzer ring time. 10-6000, unit: 100ms
		ring_interval	Buzzer ring interval. 100-10000, unit: ms

Setup:


```
{
    "msg_id": 1533,
    "device_info": {
        "mac": "4c11ae8be624"
    },
    "data": {
        "mac": "d9a5fa6f3882",
        "ring_time": 500,
        "ring_interval": 20
    }
}
```


8.36 Report Buzzer set result (533)

When the gateway gets a reply from the BXP-S device for setup Buzzer command, it will report the result to cloud.

Flag	Msg id offset	Parameter name	Parameter value
Notify	533	result_code	number. 0-3 0: command success 1: the BLE device is not connected by gateway 2: command is not supported 3: command timeout
		mac	The mac of BLE devices connected by gateway


```
{
```

    "msg_id": 3533,
    "device_info": {
        "mac": "4c11ae8be624"
    }
    "data": {
        "mac": "d9a5fa6f3882",
        "result_code": 0
    }

}

9. Connect and Manage MK PIR device

Send command from cloud to the gateway, to make the gateway connect and communicate with the MK PIR device.

9.1 Connect MK PIR device (550)

Send command from cloud to the gateway, to make the gateway connect and communicate with the MK PIR device.

Flag	Msg id offset	Parameter name	Parameter value
Write	550	mac	string. 12 characters (6 bytes of BLE device mac address)
		passwd	string. 0-16 characters (0-8 bytes of password).

Setup:

{
    "msg_id": 1550,
    "device_info": {
        "mac": "142b2fe9b79c"
    },
    "data": {
        "mac" : "d29b4edf5e84",
        "passwd" : "Moko4321"
    }
}

Reply:

{
    "msg_id": 1550,
    "device_info": {
        "mac": "142b2fe9b79c"
    },

```
    "result_code":  0,
    "result_msg":  "success"
}
```

9.2 Report connection result (551)

After the gateway finishes the connection, it will actively report the connection result to cloud. If connecting with MK PIR device successfully, the gateway will also inquire the device information from the device and transfer to cloud.

Flag	Msg id offset	Parameter name	Parameter value
Notify	551	mac	string. 12 characters (6 bytes of BLE device mac address)
		result_code	number. 0-8 0: Connect succeed 1: Gateway BLE scan disabled 2: Exceed the connectable number (the gateway is connected with an another BLE device) 3: BLE device connected (the BLE device has been connected with gateway) 4: BLE device out of range 5: BLE device un-connectable 6: Connect fail 7: BLE device type error(connection type error) 8: Password error
		result_msg	string. The description for the result code
		product_model	string. Valid when result code is 0
		company_name	string. Valid when result code is 0
		hardware_version	string. Valid when result code is 0
		software_version	string. Valid when result code is 0
		firmware_version	string. Valid when result code is 0

		run_time	Number, unit: second Valid when result code is 0
		battery_v	number. unit: mV Valid when result code is 0


```
{
  "msg_id" : 3551,
  "device_info" : {
    "mac" : "142b2fe9b79c"
  },
  "data" : {
    "mac" : "d29b4edf5e84",
    "result_code" : 0,
    "result_msg" : "Connect succeed",
    "product_model" : "MkiBeacon",
    "company_name" : "MOKO TECHNOLOGY LTD.",
    "hardware_version" : "MKBNseries",
    "software_version" : "nRF52-SDK14.2",
    "firmware_version" : "V1.0.3",
    "run_time" : 45729766,
    "battery_v" : 3594
  }
}
```


9.3 Get BLE device information (552)

Send command from cloud to the gateway, to make the gateway inquire the device information of the BLE device.

Flag	Msg id offset	Parameter name	Parameter value
Write	552	mac	string. 12 characters (6 bytes of BLE device mac address)

Setup:


```
{
  "msg_id" : 1552,
  "device_info" : {
    "mac" : "142b2fe9b79c"
  },
  "data" : {
    "mac" : "d29b4edf5e84"
  }
}
```


Reply:

```
{
  "msg_id" : 1552,
  "device_info" : {
    "mac" : "142b2fe9b79c"
  },
  "result_code" : 0,
  "result_msg" : "success"
}
```


9.4 Report BLE device information (553)

After the gateway gets device information from the BLE device, it will transfer to cloud.

Flag	Msg id offset	Parameter name	Parameter value
Notify	553	mac	string. 12 characters (6 bytes of BLE mac address)
		result_code	number. 0-2 0: success 1: the BLE device is not connected by gateway 2: command is not supported
		product_model	string. Valid when result code is 0
		company_name	string. Valid when result code is 0
		hardware_version	string. Valid when result code is 0
		software_version	string. Valid when result code is 0
		firmware_version	string. Valid when result code is 0


```
{
  "msg_id" : 3553,
  "device_info" : {
    "mac" : "142b2fe9b79c"
  },
  "data" : {
    "mac" : "d29b4edf5e84",
    "result_code" : 0,
    "product_model" : "MkiBeacon",
    "company_name" : "MOKO  TECHNOLOGY  LTD.",
```

```
        "hardware_version" : "MKBNseries",
        "software_version" : "nRF52-SDK14.2",
        "firmware_version" : "V1.0.3"
    }
}
```

9.5 Get battery level (554)

To inquire the battery status of the BLE device via gateway.

Flag	Msg id offset	Parameter name	Parameter value
Write	554	mac	string. 12 characters (6 bytes of BLE device mac address)

Setup:

```
{
    "msg_id" : 1554,
    "device_info" : {
        "mac" : "142b2fe9b79c"
    },
    "data" : {
        "mac" : "d29b4edf5e84"
    }
}
```

Reply:

```
{
    "msg_id" : 1554,
    "device_info" : {
        "mac" : "142b2fe9b79c"
    },
    "result_code" : 0,
    "result_msg" : "success"
}
```

9.6 Report battery level (555)

After the gateway get battery level from the BXP-S device, it will report to cloud.

Flag	Msg id offset	Parameter name	Parameter value
Notify	555	mac	string. 12 characters (6 bytes of Button mac address)

		result_code	number. 0-2 0: success 1: the BLE device is not connected by gateway 2: command is not supported 3: command timeout
		battery_level	number, battery percentage, unit: % Valid when the result code is 0


```
{
	"msg_id": 3555,
	"device_info": {
		"mac": "4c11ae8be624"
	},
	"data": {
		"mac": "d9a5fa6f3882",
		"result_code": 0,
		"battery_level": 89
	}
}
```


9.7 Set history sensor data notify switch(556)

The MK PIR device can notify PIR sensor history data, this command enables or disables the notify switch.

Flag	Msg id offset	Parameter name	Parameter value
Write	556	mac	string. 12 characters (6 bytes of BLE device mac address)
		switch_value	number. 0: disable 1: enable

Setup:


```
{
	"msg_id": 1556,
	"device_info": {
		"mac": "142b2fe9b79c"
	},
	"data": {
		"mac" : "d29b4edf5e84",
		"switch_value" : 1
	}
```

```
}
```
Reply:
```
{
	"msg_id": 1556,
	"device_info": {
		"mac": "142b2fe9b79c"
	},
	"result_code": 0,
	"result_msg": "success"
}
```

9.8 Report BLE device history sensor data (556)

After the gateway gets the history sensor data from the BLE device, it will report to cloud.

Flag	Msg id offset	Parameter name	Parameter value
Notify	558	mac	string. 12 characters (6 bytes of BLE mac address)
		hall_status	number. 0: close 1: open
		pir_status	number. 0: not occupied 1: occupied

```
{
  "msg_id" : 3558,
  "device_info" : {
    "mac" : "142b2fe9b79c"
  },
  "data" : {
    "mac" : "d29b4edf5e84",
    "hall_status" : 1,
    "pir_status" : 1
  }
}
```

9.9 Get PIR sensor sensitivity(559)

To get the PIR sensor sensitivity.

Flag	Msg id offset	Parameter name	Parameter value
Write	559	mac	string. 12 characters (6 bytes of BLE device mac address)

Setup:


```
{
    "msg_id": 1559,
    "device_info": {
        "mac": "142b2fe9b79c"
    },
    "data": {
        "mac" : "d29b4edf5e84"
    }
}
```


Reply:


```
{
    "msg_id": 1559,
    "device_info": {
        "mac": "142b2fe9b79c"
    },
    "result_code": 0,
    "result_msg": "success"
}
```


9.10 Report PIR sensor sensitivity(560)

After the gateway gets the PIR sensor sensitivity from the BLE device, it will report to cloud.

Flag	Msg id offset	Parameter name	Parameter value
Notify	560	mac	string. 12 characters (6 bytes of BLE mac address)
		result_code	number. 0-2 0: success 1: the BLE device is not connected by gateway 2: command is not supported 3: command timeout
		sensitivity	number. 0: low 1: medium 2: high

```
{
   "msg_id" : 3560,
   "device_info" : {
      "mac" : "142b2fe9b79c"
   },
   "data" : {
        "mac" : "d29b4edf5e84",
        "result_code" : 0,
      "sensitivity" : 0
   }
}
```

9.11 Set PIR sensor sensitivity(561)

To set the PIR sensor sensitivity.

Flag	Msg id offset	Parameter name	Parameter value
Write	561	mac	string. 12 characters (6 bytes of BLE device mac address)
		sensitivity	number. 0: low 1: medium 2: high

Setup:

```
{
      "msg_id": 1561,
      "device_info": {
            "mac": "142b2fe9b79c"
      },
      "data": {
            "mac" : "d29b4edf5e84",
            "sensitivity" : 0
      }
}
```

Reply:

```
{
      "msg_id": 1561,
      "device_info": {
            "mac": "142b2fe9b79c"
      },
      "result_code": 0,
      "result_msg": "success"
}
```

9.12 Report PIR sensor sensitivity setup result(562)

After the gateway gets a response from the BLE device for the PIR sensor sensitivity setup commands it will report to cloud.

Flag	Msg id offset	Parameter name	Parameter value
Notify	562	mac	string. 12 characters (6 bytes of BLE mac address)
		result_code	number. 0-2 0: success 1: the BLE device is not connected by gateway 2: command is not supported 3: command timeout


```
{
  "msg_id" : 3562,
  "device_info" : {
    "mac" : "142b2fe9b79c"
  },
  "data" : {
     "mac" : "d29b4edf5e84",
     "result_code" : 0,
  }
}
```


9.13 Get PIR sensor delay status(563)

To get the PIR sensor delay status.

Flag	Msg id offset	Parameter name	Parameter value
Write	563	mac	string. 12 characters (6 bytes of BLE device mac address)

Setup:


```
{
    "msg_id": 1563,
    "device_info": {
        "mac": "142b2fe9b79c"
    },
    "data": {
        "mac" : "d29b4edf5e84"
    }
}
```

Reply:

```
{
    "msg_id": 1563,
    "device_info": {
        "mac": "142b2fe9b79c"
    },
    "result_code": 0,
    "result_msg": "success"
}
```

9.14 Report PIR sensor delay status(564)

After the gateway gets the PIR sensor delay status from the BLE device, it will report to cloud.

Flag	Msg id offset	Parameter name	Parameter value
Notify	564	mac	string. 12 characters (6 bytes of BLE mac address)
		result_code	number. 0-2 0: success 1: the BLE device is not connected by gateway 2: command is not supported 3: command timeout
		delay_status	number. 0: low 1: medium 2: high

```
{
  "msg_id" : 3564,
  "device_info" : {
    "mac" : "142b2fe9b79c"
  },
  "data" : {
     "mac" : "d29b4edf5e84",
     "result_code" : 0,
    "delay_status" : 0
  }
}
```

9.15 Set PIR sensor delay status(565)

To set the PIR sensor sensitivity.

Flag	Msg id offset	Parameter name	Parameter value
Write	565	mac	string. 12 characters (6 bytes of BLE device mac address)
		delay_status	number. 0: low 1: medium 2: high

Setup:


```
{
    "msg_id": 1565,
    "device_info": {
        "mac": "142b2fe9b79c"
    },
    "data": {
        "mac" : "d29b4edf5e84",
        "delay_status" : 0
    }
}
```


Reply:


```
{
    "msg_id": 1565,
    "device_info": {
        "mac": "142b2fe9b79c"
    },
    "result_code": 0,
    "result_msg": "success"
}
```


9.16 Report PIR sensor delay setup result(562)

After the gateway gets a response from the BLE device for the PIR sensor sensitivity setup commands it will report to cloud.

Flag	Msg id offset	Parameter name	Parameter value
Notify	566	mac	string. 12 characters (6 bytes of BLE mac address)
		result_code	number. 0-2 0: success

			1: the BLE device is not connected by gateway 2: command is not supported 3: command timeout


```
{
  "msg_id" : 3566,
  "device_info" : {
    "mac" : "142b2fe9b79c"
  },
  "data" : {
     "mac" : "d29b4edf5e84",
     "result_code" : 0,
  }
}
```


9.17 Set device power off (567)

To power off the beacon via gateway.

Flag	Msg id offset	Parameter name	Parameter value
Write	567	mac	The mac of BLE devices connected by gateway

setup:


```
{
    "msg_id": 1567,
    "device_info": {
        "mac": "142b2fe9b79c"
    }
   "data": {
        "mac": "d29b4edf5e84"
    }

}
```


9.18 Report set device power off result(568)

When the gateway gets a reply from the BLE device for the power off set command, it will report the result to cloud.

Flag	Msg id offset	Parameter name	Parameter value

Notify	568	mac	string. 12 characters BLE device MAC address
		result_code	number. 0-3 0: command success 1: the BLE device is not connected by gateway 2: command is not supported 3: command timeout


```
{
      "msg_id": 3568,
      "device_info": {
            "mac": "142b2fe9b79c"
      },
      "data": {
            "mac": "d29b4edf5e84",
            "result_code": 0
      }
}
```


9.19 Get broadcast parameters (569)

To inquire Bluetooth broadcast parameters of BLE device.

Flag	Msg id offset	Parameter name	Parameter value
Write	569	mac	string. 12 characters (6 bytes of BLE device mac address)

Setup:


```
{
      "msg_id": 1569,
      "device_info": {
            "mac": "142b2fe9b79c"
      },
      "data": {
            "mac" : "d29b4edf5e84"
      }
}
```


9.20 Report broadcast parameters (570)

When the gateway gets a reply from the BLE device for the get Bluetooth broadcast parameters command, it will report to cloud.

Flag	Msg id offset	Parameter name	Parameter value
Notify	570	mac	string. 12 characters (6 bytes of BLE device mac address)
		result_code	number. 0-3 0: command success 1: the BLE device is not connected by gateway 2: command is not supported 3: command timeout
		adv_interval	number. unit: 100ms
		tx_power	number. unit: dBm 0:4dbm 1: 0dbm 2:-4dbm 3:-8dbm 4:-12dbm 5:-16dbm 6:-20dbm 7:-40dbm


```
{
  "msg_id" : 3570,
  "device_info" : {
    "mac" : "142b2fe9b79c"
  },
  "data" : {
    "mac" : "d29b4edf5e84",
    "result_code" : 0,
    "adv_interval" : 10,
    "tx_power" : 4
  }
}
```


9.21 Set broadcast parameters (571)

To set beacon Bluetooth broadcast parameters for BLE device via gateway.

Flag	Msg id offset	Parameter name	Parameter value
Write	571	mac	string. 12 characters (6 bytes of Button mac address)
		adv_interval	number. 1-100 unit:100ms
		tx_power	number. unit: dBm 0:4dbm

			1: 0dbm 2:-4dbm 3:-8dbm 4:-12dbm 5:-16dbm 6:-20dbm 7:-40dbm

Setup:


```
{
	"msg_id": 1571,
	"device_info": {
		"mac": "142b2fe9b79c"
	},
	"data": {
		"mac": "d29b4edf5e84",
		"adv_interval": 20,
		"tx_power": 4
		  }
}
```


9.22 Report broadcast parameters set result (572)

When the gateway gets reply from the BLE device for the Bluetooth broadcast parameters set command, it will report the result to cloud.

Flag	Msg id offset	Parameter name	Parameter value
Notify	572	result_code	number. 0-3 0: command success 1: the BLE device is not connected by gateway 2: command is not supported 3: command timeout
		mac	string. 12 characters (6 bytes of BLE device mac address)


```
{
  "msg_id" : 3572,
  "device_info" : {
    "mac" : "142b2fe9b79c"
  },
  "data" : {
    "mac" : "d29b4edf5e84",
    "result_code" : 0
  }
```

```
}
```

10. Connect and Manage MK TOF device

Send command from cloud to the gateway, to make the gateway connect and communicate with the MK TOF device.

10.1 Connect MK PIR device (600)

Send command from cloud to the gateway, to make the gateway connect and communicate with the MK TOF device.

Flag	Msg id offset	Parameter name	Parameter value
Write	600	mac	string. 12 characters (6 bytes of BLE device mac address)
		passwd	string. 0-16 characters (0-8 bytes of password).

Setup:


```
{
	"msg_id": 1600,
	"device_info": {
		"mac": "142b2fe9b79c"
	},
	"data": {
		"mac" : "df442c64b527",
		"passwd" : "Moko4321"
	}
}
```


Reply:


```
{
	"msg_id": 1600,
	"device_info": {
		"mac": "142b2fe9b79c"
	},
	"result_code": 0,
	"result_msg": "success"
}
```

10.2 Report connection result (601)

After the gateway finishes the connection, it will actively report the connection result to cloud. If connecting with the BLE device successfully, the gateway will also inquire the device information from the device and transfer to cloud.

Flag	Msg id offset	Parameter name	Parameter value
Notify	601	mac	string. 12 characters (6 bytes of BLE device mac address)
		result_code	number. 0-8 0: Connect succeed 1: Gateway BLE scan disabled 2: Exceed the connectable number (the gateway is connected with an another BLE device) 3: BLE device connected (the BLE device has been connected with gateway) 4: BLE device out of range 5: BLE device un-connectable 6: Connect fail 7: BLE device type error(connection type error) 8: Password error
		result_msg	string. The description for the result code
		product_model	string. Valid when result code is 0
		company_name	string. Valid when result code is 0
		hardware_version	string. Valid when result code is 0
		software_version	string. Valid when result code is 0
		firmware_version	string. Valid when result code is 0
		battery_v	number. unit: mV Valid when result code is 0


```
{
   "msg_id" : 3601,
   "device_info" : {
```

```
    "mac" : "142b2fe9b79c"
  },
  "data" : {
    "mac" : "df442c64b527",
    "result_code" : 0,
    "result_msg" : "Connect succeed",
    "product_model" : "MkiBeacon",
    "company_name" : "MOKO TECHNOLOGY LTD.",
    "hardware_version" : "MKBNseries",
    "software_version" : "nRF52-SDK14.2",
    "firmware_version" : "MK_ToF_Sensor V1.1.2",
    "battery_v" : 3030
  }
}
```


10.3 Get BLE device information (602)

Send command from cloud to the gateway, to make the gateway inquire the device information of the BLE device.

Flag	Msg id offset	Parameter name	Parameter value
Write	602	mac	string. 12 characters (6 bytes of BLE device mac address)

Setup:


```
{
  "msg_id" : 1602,
  "device_info" : {
    "mac" : "142b2fe9b79c"
  },
  "data" : {
    "mac" : "df442c64b527"
  }
}
```


Reply:


```
{
  "msg_id" : 1602,
  "device_info" : {
    "mac" : "142b2fe9b79c"
  },
  "result_code" : 0,
  "result_msg" : "success"
}
```

10.4 Report BLE device information (603)

After the gateway gets device information from the BLE device, it will transfer to cloud.

Flag	Msg id offset	Parameter name	Parameter value
Notify	603	mac	string. 12 characters (6 bytes of BLE mac address)
		result_code	number. 0-2 0: success 1: the BLE device is not connected by gateway 2: command is not supported
		product_model	string. Valid when result code is 0
		company_name	string. Valid when result code is 0
		hardware_version	string. Valid when result code is 0
		software_version	string. Valid when result code is 0
		firmware_version	string. Valid when result code is 0


```
{
  "msg_id" : 3603,
  "device_info" : {
    "mac" : "142b2fe9b79c"
  },
  "data" : {
    "mac" : "df442c64b527",
    "result_code" : 0,
    "product_model" : "MkiBeacon",
    "company_name" : "MOKO TECHNOLOGY LTD.",
    "hardware_version" : "MKBNseries",
    "software_version" : "nRF52-SDK14.2",
    "firmware_version" : "MK_ToF_Sensor V1.1.2"
  }
}
```

10.5 Get battery level (604)

To inquire the battery status of the BLE device via gateway.

Flag	Msg id offset	Parameter name	Parameter value
Write	604	mac	string. 12 characters (6 bytes of BLE device mac address)

Setup:


```
{
   "msg_id" : 1604,
   "device_info" : {
      "mac" : "142b2fe9b79c"
   },
   "data" : {
      "mac" : "df442c64b527"
   }
}
```


Reply:


```
{
   "msg_id" : 1604,
   "device_info" : {
      "mac" : "142b2fe9b79c"
   },
   "result_code" : 0,
   "result_msg" : "success"
}
```


10.6 Report battery level (605)

After the gateway get battery level from the BXP-S device, it will report to cloud.

Flag	Msg id offset	Parameter name	Parameter value
Notify	605	mac	string. 12 characters (6 bytes of Button mac address)
		result_code	number. 0-2 0: success 1: the BLE device is not connected by gateway 2: command is not supported 3: command timeout
		battery_V	number, battery voltage unit: mV

			Valid when the result code is 0


```
{
  "msg_id" : 3605,
  "device_info" : {
    "mac" : "142b2fe9b79c"
  },
  "data" : {
    "mac" : "df442c64b527",
    "result_code" : 0,
    "battery_v" : 3030
  }
}
```


10.7 Set 3-axis sensor data notify switch(606)

The MK TOF device can notify real time 3-axis sensor data, this command enables or disables the notify switch.

Flag	Msg id offset	Parameter name	Parameter value
Write	606	mac	string. 12 characters (6 bytes of BLE device mac address)
		switch_value	number. 0: disable 1: enable

Setup:


```
{
    "msg_id": 1606,
    "device_info": {
        "mac": "142b2fe9b79c"
    },
    "data": {
        "mac" : "df442c64b527",
        "switch_value" : 1
    }
}
```


Reply:


```
{
    "msg_id": 1606,
    "device_info": {
        "mac": "142b2fe9b79c"
    },
    "result_code": 0,
```

```
    "result_msg": "success"
}
```

10.8 Report 3-axis data notify switch setup result(607)

After the gateway get a response for the notify switch setup command, it will report the result to cloud.

Flag	Msg id offset	Parameter name	Parameter value
Notify	607	mac	string. 12 characters BLE device MAC address
		result_code	number. 0-3 0: command success 1: the BLE device is not connected with gateway 2: command is not supported 3: command timeout

```
{
  "msg_id" : 3607,
  "device_info" : {
    "mac" : "142b2fe9b79c"
  },
  "data" : {
    "mac" : "df442c64b527",
    "result_code" : 1
  }
}
```

10.9 Report 3-axis data (608)

When the BLE device enables the 3-axis data notify switch, the gateway will receive real time 3-axis data from the BLE device, then gateway will transfer to cloud.

Flag	Msg id offset	Parameter name	Parameter value
Notify	608	mac	string. 12 characters BLE device MAC address
		x_axis_data	number,mg
		y_axis_data	number,mg
		z_axis_data	number,mg

```
{
    "msg_id": 3608,
    "device_info": {
        "mac": "142b2fe9b79c"
    },
    "data": {
        "mac": "df442c64b527",
        "x_axis_data": 0,
        "y_axis_data": 50,
        "z_axis_data": 1500
    }
}
```

10.10 Set device power off (609)

To power off the beacon via gateway.

Flag	Msg id offset	Parameter name	Parameter value
Write	609	mac	The mac of BLE devices connected by gateway

setup:

```
{
    "msg_id": 1609
    "device_info": {
        "mac": "142b2fe9b79c"
    }
   "data": {
        "mac": "df442c64b527"
    }

}
```

10.11 Report set device power off result(610)

When the gateway gets a reply from the BLE device for the power off set command, it will report the result to cloud.

Flag	Msg id offset	Parameter name	Parameter value
Notify	610	mac	string. 12 characters BLE device MAC address

		result_code	number. 0-3 0: command success 1: the BLE device is not connected by gateway 2: command is not supported 3: command timeout


```
{
    "msg_id": 3610
    "device_info": {
        "mac": "142b2fe9b79c"
    },
    "data": {
        "mac": "df442c64b527",
        "result_code": 0
    }
}
```


10.12 Get broadcast parameters (611)

To inquire Bluetooth broadcast parameters of BLE device.

Flag	Msg id offset	Parameter name	Parameter value
Write	611	mac	string. 12 characters (6 bytes of BLE device mac address)

Setup:


```
{
    "msg_id": 1611,
    "device_info": {
        "mac": "142b2fe9b79c"
    },
    "data": {
        "mac" : "df442c64b527"
    }
}
```


10.13 Report broadcast parameters (612)

When the gateway gets a reply from the BLE device for the get Bluetooth broadcast parameters command, it will report to cloud.

Flag	Msg id offset	Parameter name	Parameter value

Notify	612	mac	string. 12 characters (6 bytes of BLE device mac address)
		result_code	number. 0-3 0: command success 1: the BLE device is not connected by gateway 2: command is not supported 3: command timeout
		adv_interval	number. 1-86400, second
		tx_power	number. unit: dBm 0:4dbm 1: 0dbm 2:-4dbm 3:-8dbm 4:-12dbm 5:-16dbm 6:-20dbm 7:-40dbm


```
{
   "msg_id" : 3612,
   "device_info" : {
      "mac" : "142b2fe9b79c"
   },
   "data" : {
      "mac" : "df442c64b527",
      "result_code" : 0,
      "adv_interval" : 10,
      "tx_power" : 4
   }
}
```


10.14 Set broadcast parameters (613)

To set beacon Bluetooth broadcast parameters for BLE device via gateway.

Flag	Msg id offset	Parameter name	Parameter value
Write	613	mac	string. 12 characters (6 bytes of Button mac address)
		adv_interval	number. 1-86400, second
		tx_power	number. unit: dBm 0:4dbm 1: 0dbm

			2:-4dbm 3:-8dbm 4:-12dbm 5:-16dbm 6:-20dbm 7:-40dbm

Setup:


```
{
    "msg_id": 1613,
    "device_info": {
        "mac": "142b2fe9b79c"
    },
    "data": {
        "mac": "df442c64b527",
        "adv_interval": 20,
        "tx_power": 4
          }
}
```


10.15 Report broadcast parameters set result (614)

When the gateway gets reply from the BLE device for the Bluetooth broadcast parameters set command, it will report the result to cloud.

Flag	Msg id offset	Parameter name	Parameter value
Notify	614	result_code	number. 0-3 0: command success 1: the BLE device is not connected by gateway 2: command is not supported 3: command timeout
		mac	string. 12 characters (6 bytes of BLE device mac address)


```
{
  "msg_id" : 3614,
  "device_info" : {
    "mac" : "142b2fe9b79c"
  },
  "data" : {
    "mac" : "df442c64b527",
    "result_code" : 0
  }
}
```

10.16 Get TOF sensor sampling parameters(615)

To get the TOF sensor parameters.

Flag	Msg id offset	Parameter name	Parameter value
Write	615	mac	string. 12 characters (6 bytes of BLE device mac address)

Setup:


```
{
    "msg_id": 1615,
    "device_info": {
        "mac": "142b2fe9b79c"
    },
    "data": {
        "mac" : "df442c64b527"
    }
}
```


Reply:


```
{
    "msg_id": 1615,
    "device_info": {
        "mac": "142b2fe9b79c"
    },
    "result_code": 0,
    "result_msg": "success"
}
```


10.17 Report TOF sensor sampling parameters(616)

After the gateway gets the TOF sensor parameters from the BLE device, it will report to cloud.

Flag	Msg id offset	Parameter name	Parameter value
Notify	616	mac	string. 12 characters (6 bytes of BLE mac address)
		result_code	number. 0-2 0: success 1: the BLE device is not connected by gateway 2: command is not supported 3: command timeout
		interval	number. Sample interval.

			1-86400, second
		count	number, sample count. 2-255
		time	Number, each sample time. 8-140,ms


```
{
  "msg_id" : 3616,
  "device_info" : {
    "mac" : "142b2fe9b79c"
  },
  "data" : {
    "mac" : "df442c64b527",
    "result_code" : 0,
    "interval" : 5,
    "count" : 5,
    "time" : 32
  }
}
```


10.18 Set TOF sensor sampling parameters(617)

After the gateway gets the TOF sensor parameters from the BLE device, it will report to cloud.

Flag	Msg id offset	Parameter name	Parameter value
Write	617	mac	string. 12 characters (6 bytes of BLE mac address)
		interval	number. Sample interval. 1-86400, second
		count	number, sample count. 2-255
		time	Number, each sample time. 8-140,ms


```
{
  "msg_id": 1617,
   "device_info" : {
    "mac" : "142b2fe9b79c"
  },
  "data" : {
    "mac" : "df442c64b527",
    "interval" : 5,
    "count" : 5,
```

"time" : 32
  }
}

Reply:

{
    "msg_id": 1617,
    "device_info": {
        "mac": "142b2fe9b79c"
    },
    "result_code": 0,
    "result_msg": "success"
}

10.19 Report TOF sensor sampling parameters setup result(618)

After the gateway gets response for the TOF sensor parameters setup command, from the BLE device, it will report to cloud.

Flag	Msg id offset	Parameter name	Parameter value
Notify	618	mac	string. 12 characters (6 bytes of BLE mac address)
		result_code	number. 0-2 0: success 1: the BLE device is not connected by gateway 2: command is not supported 3: command timeout

{
  "msg_id" : 3618,
  "device_info" : {
    "mac" : "142b2fe9b79c"
  },
  "data" : {
    "mac" : "df442c64b527",
    "result_code" : 0,
  }
}

10.20 Set TOF distance mode (619)

To get the TOF sensor distance mode.

Flag	Msg id offset	Parameter name	Parameter value
Write	619	mac	string. 12 characters (6 bytes of BLE device mac address)

Setup:


```
{
    "msg_id": 1619,
    "device_info": {
        "mac": "142b2fe9b79c"
    },
    "data": {
        "mac" : "df442c64b527"
    }
}
```


Reply:


```
{
    "msg_id": 1619,
    "device_info": {
        "mac": "142b2fe9b79c"
    },
    "result_code": 0,
    "result_msg": "success"
}
```


10.21 Report TOF distance mode (620)

After the gateway gets the response from the BLE device for the TOF distance mode enquiry command, it will report to cloud.

Flag	Msg id offset	Parameter name	Parameter value
Notify	620	mac	string. 12 characters (6 bytes of BLE mac address)
		result_code	number. 0-2 0: success 1: the BLE device is not connected by gateway 2: command is not supported 3: command timeout
		mode	number

			0: short distance 1: long distance


```
{
   "msg_id" : 3620,
   "device_info" : {
      "mac" : "142b2fe9b79c"
   },
   "data" : {
      "mac" : "df442c64b527",
      "result_code" : 0,
      "mode" : 0
   }
}
```


10.22 Set TOF distance mode(621)

Set the TOF distance mode via gateway.

Flag	Msg id offset	Parameter name	Parameter value
Write	621	mac	string. 12 characters (6 bytes of BLE mac address)
		mode	number 0: short distance 1: long distance


```
{
   "msg_id": 1621,
    "device_info" : {
      "mac" : "142b2fe9b79c"
   },
   "data" : {
      "mac" : "df442c64b527",
      "mode" : 1
   }
}
```


Reply:


```
{
      "msg_id": 1621,
      "device_info": {
            "mac": "142b2fe9b79c"
      },
```

```
    "result_code": 0,
    "result_msg": "success"
}
```


10.23 Report TOF distance mode setup result(622)

After the gateway gets response for the TOF sensor distance mode setup command, from the BLE device, it will report to cloud.

Flag	Msg id offset	Parameter name	Parameter value
Notify	622	mac	string. 12 characters (6 bytes of BLE mac address)
		result_code	number. 0-2 0: success 1: the BLE device is not connected by gateway 2: command is not supported 3: command timeout


```
{
  "msg_id" : 3622,
  "device_info" : {
    "mac" : "142b2fe9b79c"
  },
  "data" : {
    "mac" : "df442c64b527",
    "result_code" : 0,
  }
}
```


10.24 Set TOF sensor data notify switch(623)

The MK TOF device can notify real time TOF sensor data, this command enables or disables the notify switch.

Flag	Msg id offset	Parameter name	Parameter value
Write	623	mac	string. 12 characters (6 bytes of BLE device mac address)
		switch_value	number. 0: disable 1: enable

Setup:


```
{
	"msg_id": 1623,
	"device_info": {
		"mac": "142b2fe9b79c"
	},
	"data": {
		"mac" : "df442c64b527",
		"switch_value" : 1
	}
}
```


Reply:


```
{
	"msg_id": 1623,
	"device_info": {
		"mac": "142b2fe9b79c"
	},
	"result_code": 0,
	"result_msg": "success"
}
```


10.25 Report TOF sensor data notify switch setup result(624)

After the gateway get a response for the notify switch setup command, it will report the result to cloud.

Flag	Msg id offset	Parameter name	Parameter value
Notify	624	mac	string. 12 characters BLE device MAC address
		result_code	number. 0-3 0: command success 1: the BLE device is not connected with gateway 2: command is not supported 3: command timeout


```
{
  "msg_id" : 3624,
  "device_info" : {
    "mac" : "142b2fe9b79c"
  },
```

```
  "data" : {
    "mac" : "df442c64b527",
    "result_code" : 1
  }
}
```

10.26 Report TOF sensor data (625)

When the BLE device enables the TOF sensor data notify switch, the gateway will receive real time TOF sensor data from the BLE device, then gateway will transfer to cloud.

Flag	Msg id offset	Parameter name	Parameter value
Notify	625	mac	string. 12 characters BLE device MAC address
		distance	number, unit: mm

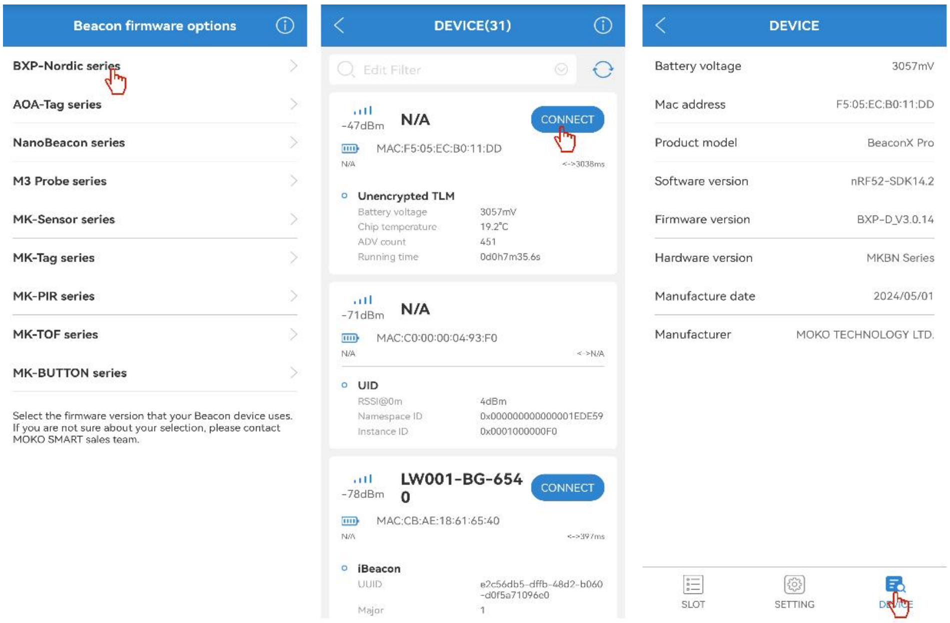
```
{
  "msg_id" : 3625,
  "device_info" : {
    "mac" : "142b2fe9b79c"
  },
  "data" : {
    "mac" : "df442c64b527",
    "distance" : 631
  }
}
```

Appendix

1. Check firmware type of your Beacon device

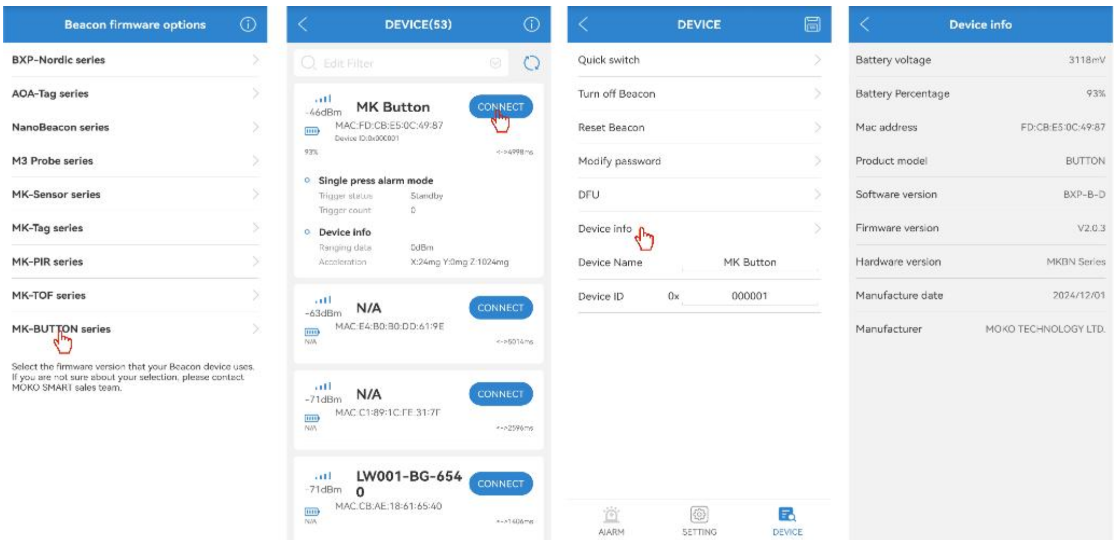
BXP-Nordic series:

1. Open the MOKO BeaconX Pro mobile APP, search the device and connect it.
2. After the device is connected, device information will be displayed in the DEVICE page, the firmware version shows the firmware type (BXP-D or BXP-C).



MK-Button series

1. Open the MOKO BeaconX Pro mobile APP, search the device and connect it.
2. After the device is connected, go to DEVICE page, then inquire the device information, the software version shows the firmware type (BXP-B-D or BXP-B-CR).



Revision History

Revision	Change notes	Editor	Date
V2.2	Initial version. Compatible with MOKO plug gateway and POE gateway V2.X.X firmware.	Weiguifen	2026.1.27
V2.3	1.Add chapter *9 Connect and Manage MK PIR device* and chapter *10 Connect and Manage MK TOF device* 2.Add battery level (percentage) info in *3.2 Report connection result* and *3.6 Report BLE device status* 3.Add battery level (percentage) info in *4.2 Report connection result* and *4.6 Report BLE device information* 4.Add battery voltage info in *8.2 Report connection result* and *8.6 Report BLE device battery info* 5. Add command to setup buzzer of BXP-S device, see chapter *8.35 Setup buzzer* and *8.36 Report buzzer set result*	Weiguifen	2026.6.25

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